VI. Experimental Evaluation, Results, and Discussion

1. Experimental Evidence Overview

This chapter reports the verified outcomes of the two computer-vision tasks defined in Chapter IV. The classification branch evaluated a four-class, single-label problem in the fixed class order Healthy, BG, WSSV, and WSSV_BG. The retained evidence covers candidate-model screening, the clean comparison between the YOLO26m-cls cross-entropy (CE) reference and the final ASL-LDAM + SimAM-DCFR configuration, controlled synthetic corruption tests, additional-source experiments, qualitative explainability, and a bounded Android deployment observation. The instance-segmentation branch evaluated BG and WSSV disease-region masks under specimen-aware protocols, with separate analyses of split leakage, baseline families, attention configurations, Fourier preprocessing, corruption sensitivity, and Healthy negative-image behavior.

Machine-readable values preserved in the final evidence were used directly where available. Unsupported row-level values were not reconstructed from memory or estimated from plots. Results are reported on a 0–100 percentage scale unless otherwise stated, and absolute changes are expressed in percentage points.

2. Dataset Composition and Evaluation Partitions

2.1 Classification Dataset Composition

The primary classification source is SDI-4, the 1,149-image ShrimpDiseaseImageBD-4 primary dataset containing Healthy, BG, WSSV, and WSSV_BG. EXT-3-Original denotes the separately mapped external ShrimpDB source with three trained classes: Healthy, BG, and WSSV. SDI-4 + EXT-3-Original denotes the four-class partition-wise integrated regime created by combining corresponding source-specific training, validation, and test partitions; WSSV_BG is contributed only by SDI-4.

Table 6.1: Classification dataset regimes and fixed partition counts

Dataset regime	Classes	Train	Validation	Test	Total
SDI-4	4	804	172	173	1,149
EXT-3-Original	3	221	47	47	315
SDI-4 + EXT-3-Original	4	1,025	219	220	1,464

Each source was split independently before partition-wise integration. The retained multi-source results are retraining experiments on the stated dataset regimes, not direct inference by the original SDI-4 checkpoint on an unseen source.

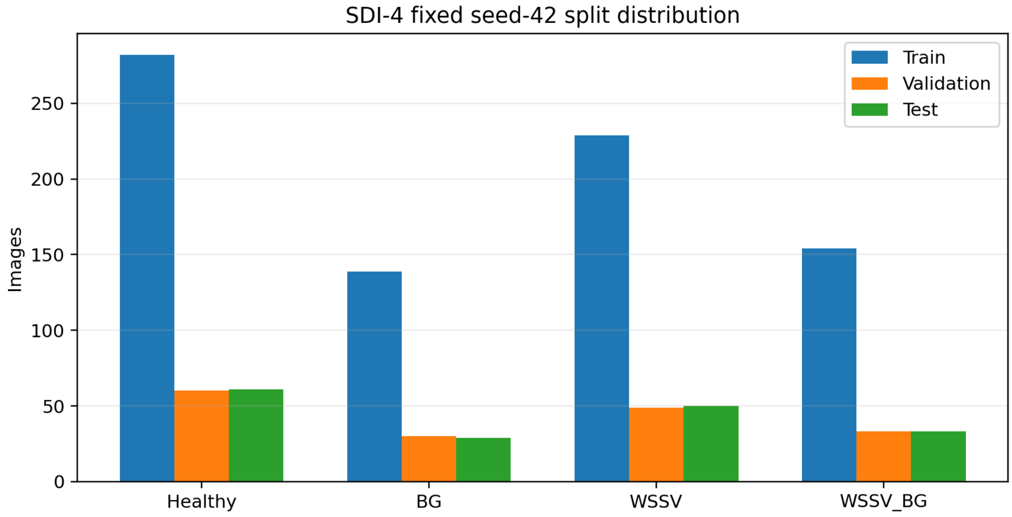

Figure 6.1: Class distribution across the fixed seed-42 SDI-4 training, validation, and test partitions.

2.2 Instance-Segmentation Dataset Composition

The recognized specimen-group subset contained 1,149 images associated with 416 filename-derived specimens. Project-authored BG or WSSV disease-region labels were available for 746 images, while 403 Healthy images were retained as empty-label negatives containing no positive disease mask. Healthy is therefore an image-status condition used during training and evaluation, not a third segmentation-mask class.

Table 6.2A. Recognized specimen-group subset under the 80/10/10 protocol

Split	Images	Specimens	Labeled disease images	Healthy empty-label images
Train	905	331	584	321
Validation	115	40	74	41
Test	129	45	88	41
Total	**1,149**	**416**	**746**	**403**

The expanded image inventory retained all 1,149 recognized specimen-group images and added 303 unmatched images. The added images were allocated by the retained project logic because compatible specimen identifiers were unavailable.

Table 6.2B. Expanded instance-segmentation image inventory

Split	Images	Recognized specimen-group images	Added unmatched images
Train	1,147	905	242
Validation	144	115	29
Test	161	129	32
Total	**1,452**	**1,149**	**303**

2.3 External and Partition-Wise Integrated Classification Datasets

EXT-3-Original maps the source folders Tom_BT, Den_Mang, and Dom_Trang to Healthy, BG, and WSSV, respectively. Its fixed partition contains 221 training images, 47 validation images, and 47 test images. The suffix "-3" identifies the three trained classes and distinguishes this project-defined experiment identifier from the official source name ShrimpDB.

SDI-4 + EXT-3-Original is the four-class partition-wise integrated regime. SDI-4 and the external source were split independently, after which train was integrated with train, validation with validation, and test with test. The resulting partition contains 1,025 training images, 219 validation images, and 220 test images. WSSV_BG remains available only from SDI-4. Because source composition, class priors, and acquisition backgrounds change, these results qualify the primary finding rather than establishing universal generalization.

3. Classification Baseline Screening

3.1 ShrimpXNet-Oriented Reference Context

ShrimpXNet was retained as the closest literature-oriented classification reference because its pipeline included background removal and ConvNeXt-Tiny-style transfer learning. It was not the architecture modified in this thesis. The team applied its own fixed seed-42 screening protocol, with cross-entropy serving as the controlled objective during candidate screening and reference-model training.

3.2 Complete TIMM Candidate Benchmark

The revised lightweight selection preference is a deployment-oriented budget of approximately 15 million parameters or fewer. Historical screening artifacts also contain larger diagnostic references; those rows are retained only when they help explain capacity behavior and are not treated as eligible final mobile candidates. This distinction is important for the YOLO26m-cls versus YOLO26x-cls comparison: YOLO26x-cls is useful as a higher-capacity diagnostic control, but its 28.36M parameters place it outside the revised deployment preference.

The retained TIMM screening and Ultralytics screening used the same fixed seed-42 classification partition. Within the deployable range, the objective was not to minimize parameter count alone, but to find a stable predictive-efficiency trade-off using Macro-F1 as the principal outcome together with accuracy, model complexity, export feasibility, and measured runtime metadata.

The complete retained seed-42 TIMM screening ledger is reported in Tables 6.3A and 6.3B on the following pages. Predictive metrics and model complexity are separated from timing metadata so that all 17 retained candidates remain readable at the thesis table font size. The uploaded CSV was cross-checked against the retained screening image before typesetting; no missing TIMM row required numerical imputation.

3.2 Complete TIMM Candidate Benchmark (continued)

Table 6.3A reports the complete predictive and complexity ledger from the retained TIMM baseline screen. The archived fractional F1 and accuracy values are displayed as percentages for consistency with the other classification tables in this chapter. Candidate order follows the retained screening rank.

Table 6.3A. Complete TIMM seed-42 predictive and complexity screening results

Model	Params (M)	Val F1 (%)	Test Accuracy (%)	Test Macro-F1 (%)
convnext_tiny_in22k	27.82	85.30	85.22	84.16
mobilenet_v3_large	4.21	79.27	82.61	80.10
efficientnet_b0	4.01	83.20	79.13	77.01
repvgg_a0	7.83	78.49	77.39	76.88
efficientnet_v2_s	20.18	78.27	77.39	76.78
shufflenet_v2_x1_0	1.26	76.17	74.78	71.85
fastvit_t8	3.26	78.29	73.91	71.03
edgenext_xx_small	1.16	71.46	72.17	70.63
mobileone_s0	4.27	73.34	71.30	69.69
mobilevit_s	4.94	80.78	73.91	69.18
mobilenetv4_conv_small	2.50	75.65	67.83	66.26
mnasnet_100	3.11	70.79	66.96	62.48
ghostnetv2_100	4.88	72.40	61.74	61.01
rexnet_100	3.52	63.99	61.74	58.55
squeezenet1_1	0.72	68.86	59.13	55.04
mobilenetv4_hybrid_medium	9.80	59.43	60.87	53.83
efficientvit_m1	2.79	68.10	57.39	52.83

ConvNeXt-Tiny-IN22K produced the strongest retained TIMM test Macro-F1 (84.16%) and test accuracy (85.22%), but its 27.82M parameters place it outside the project's approximately 15M-parameter deployment preference. Within that preference, MobileNetV3-Large was the strongest retained TIMM row by test Macro-F1 (80.10%) at 4.21M parameters. These results show that the TIMM screen supplied useful lightweight references, but none matched the later YOLO26m-cls reference on the locked primary screening metric.

3.2 Complete TIMM Candidate Benchmark (continued)

Table 6.3B preserves the speed fields associated with the same 17 TIMM candidates. Because the retained screening artifact does not independently document an identical timing harness for the later Ultralytics measurements, these values are treated as within-TIMM auxiliary screening metadata rather than as a direct cross-framework latency benchmark.

Table 6.3B. Complete TIMM seed-42 training and inference timing ledger

Model	Training Time (s)	Reported Inference Time (s)	FPS	Latency (ms)
convnext_tiny_in22k	152.0	2.34	49.2	20.34
mobilenet_v3_large	234.1	2.52	45.6	21.93
efficientnet_b0	235.3	2.47	46.5	21.52
repvgg_a0	129.3	2.39	48.1	20.81
efficientnet_v2_s	195.4	3.57	32.2	31.06
shufflenet_v2_x1_0	226.0	2.51	45.7	21.86
fastvit_t8	110.1	2.56	44.8	22.30
edgenext_xx_small	181.3	2.39	48.2	20.75
mobileone_s0	136.6	4.11	28.0	35.76
mobilevit_s	150.7	2.47	46.5	21.51
mobilenetv4_conv_small	178.1	2.38	48.2	20.74
mnasnet_100	180.4	2.39	48.2	20.76
ghostnetv2_100	92.3	3.30	34.8	28.71
rexnet_100	129.2	2.54	45.3	22.06
squeezenet1_1	135.5	2.26	50.8	19.67
mobilenetv4_hybrid_medium	147.8	3.38	34.0	29.43
efficientvit_m1	148.9	3.17	36.3	27.55

The timing ledger also demonstrates why speed was not used as a stand-alone selection rule. SqueezeNet1_1 recorded the highest retained TIMM throughput (50.8 FPS; 19.67 ms), but its test Macro-F1 was only 55.04%. Conversely, several stronger predictive models remained near the 20-22 ms range. The screening decision therefore considered predictive quality and efficiency jointly.

3.3 Ultralytics Classification Benchmark

The Ultralytics screening covered classification candidates from the YOLOv8, YOLO11, and YOLO26 families. Jointly traceable performance and efficiency metadata were retained for five representative rows. YOLO26m-cls was selected because it provided the strongest retained Macro-F1 among the practically sized candidates while remaining close to the revised 15M-parameter deployment preference. YOLO26x-cls is retained in Table 6.3 as an explicitly out-of-budget capacity diagnostic rather than as a deployment candidate.

3.4 Reference-Model Selection

Reference-model selection considered predictive performance, parameter count, model size, latency, and FPS. The timing values in Table 6.3 are auxiliary GPU screening measurements and are not Android latency. They must not be reinterpreted as measurements from the final dual-T4 training protocol unless matching provenance exists. The selected YOLO26m-cls reference was subsequently used for the controlled CE comparison and the final ASL-LDAM + SimAM-DCFR integration.

Table 6.3: Verified representative Ultralytics selection summary

Model	Accuracy (%)	Macro-F1 (%)	Params (M)	Size (MB)	Latency (ms)	FPS
YOLO26m-cls	**89.02**	**89.02**	10.36	19.92	3.16	316
YOLO26x-cls	**89.02**	88.13	28.36	54.37	4.59	218
YOLOv8m-cls	87.86	87.18	15.78	30.22	2.44	410
YOLOv8s-cls	87.86	87.10	**5.09**	**9.79**	**1.87**	**535**
YOLO11m-cls	87.28	86.75	10.36	19.92	3.16	317

Within this representative set, YOLO26m-cls produced the highest Macro-F1 while remaining below the approximate 15-million-parameter preference. YOLO26x-cls matched its accuracy but had lower Macro-F1 and substantially greater size. YOLOv8s-cls was faster and smaller in the auxiliary screening measurement, but its Macro-F1 was lower. These trade-offs support the engineering selection of YOLO26m-cls within the evaluated candidate set; they do not establish universal superiority over untested architectures.

A separate train/validation diagnostic was retained specifically to test whether the additional capacity of YOLO26x-cls improved optimization or generalization under the same seed-42 protocol. No new test inference was performed for this diagnostic. YOLO26m-cls reached a best validation Top-1 accuracy of 84.30% at epoch 25 and finished at 82.56%, whereas YOLO26x-cls reached 80.23% at epoch 29 and finished at the same 80.23%. Final validation loss was also lower for YOLO26m-cls (0.4134) than for YOLO26x-cls (0.4749). Both runs show some train–validation separation, but the evidence does not justify claiming that YOLO26x alone “overfit.” The defensible conclusion is narrower: extra capacity did not improve validation performance or the historical test Macro-F1 under the fixed protocol.

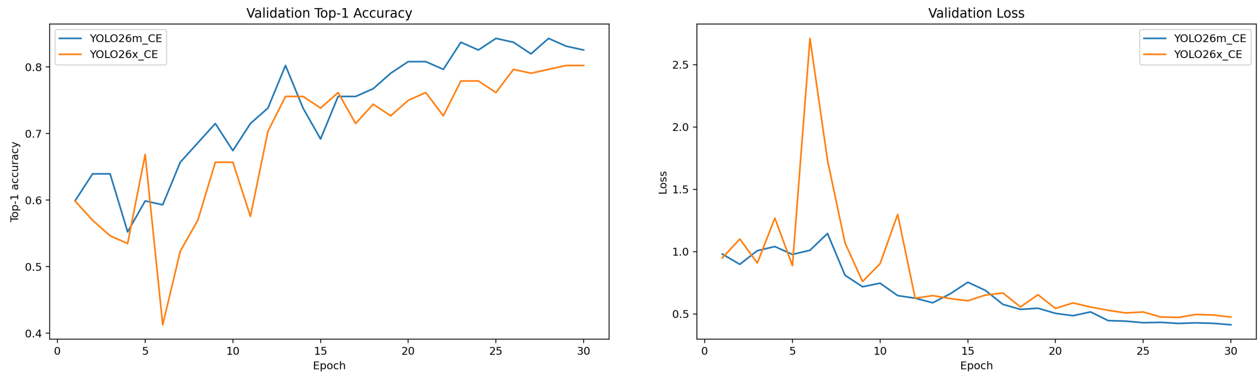

Figure 6.1A. YOLO26m-cls versus YOLO26x-cls validation diagnostics under the same seed-42 CE protocol. Left: validation Top-1 accuracy by epoch. Right: validation loss by epoch. The larger YOLO26x-cls model did not improve the validation trajectory and is outside the revised approximately 15M-parameter deployment preference. These curves are diagnostic training/validation evidence and do not replace the locked primary clean-test result.

The historical screen used test-partition Macro-F1 as a selection signal. Repeated inspection of that partition can bias model-family ranking. This chapter preserves the actual protocol rather than relabeling it as validation-only selection, and it does not claim multi-seed significance.

4. Main Classification Performance

4.1 CE Baseline versus Final Proposed Model

The primary comparison used the same YOLO26m-cls reference architecture, fixed seed-42 split, background-removed representation, optimizer family, schedule, epoch budget, and evaluation definitions. Cross-entropy defined the reference condition. The final configuration replaced the training objective with ASL-LDAM and inserted SimAM-DCFR before the classification head.

Table 6.4: Main clean-test classification comparison

Method	**Accuracy (%)**	**Macro-F1 (%)**	**Kappa $\times$100**	**Δ Macro-F1 (pp)**
YOLO26m-cls + CE	89.02	89.02	85.05	0.00
YOLO26m-cls + ASL-LDAM + SimAM-DCFR	**91.33**	**91.01**	**88.16**	**+1.99**

The final model increased Macro-F1 by 1.99 percentage points and accuracy by 2.31 points. Cohen's Kappa increased from 0.8505 to 0.881593, corresponding to approximately +3.11 points on the displayed $\times$100 scale. Under the recorded fixed-seed protocol, the improvement indicates a better balance of classwise predictions, but it is not a confidence interval or a statistical-significance result.

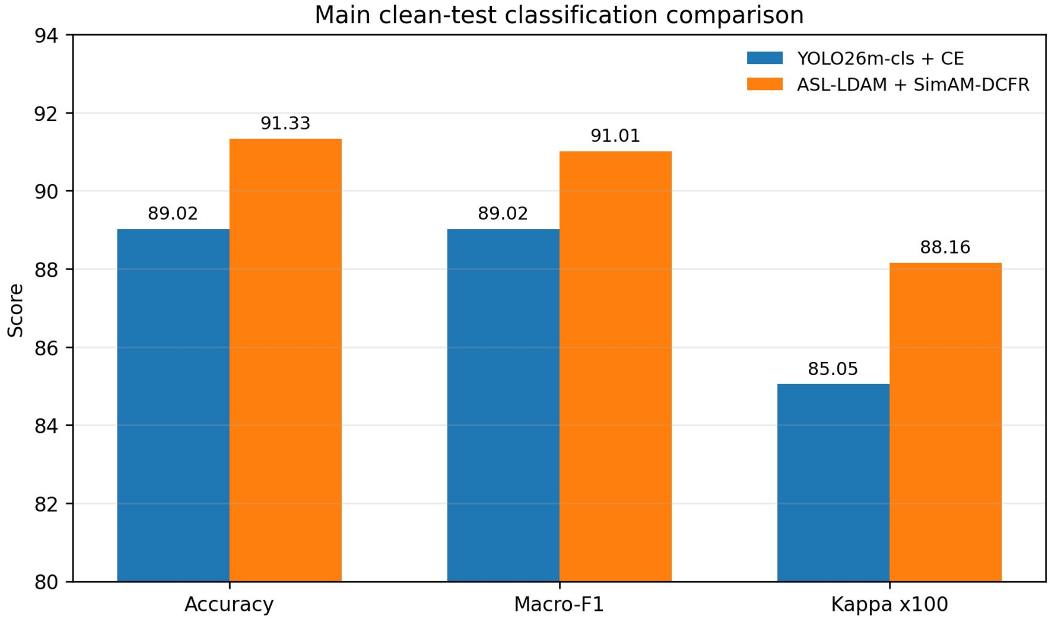

Figure 6.2: Accuracy, Macro-F1, and Cohen's Kappa for the clean CE reference and final proposed classifier.

Figure 6.2 makes the scale of the improvement visible without implying that the three metrics are interchangeable. Macro-F1 remains the primary outcome because it is less dominated by the majority class than raw accuracy.

4.2 Classwise Performance

The verified final artifacts do not contain the prediction-level file or complete classwise report for the 173-image primary test partition. Exact primary-test precision, recall, F1, support, and transition counts are therefore not reconstructed. This is a material reporting limitation: an overall Macro-F1 increase does not identify which classes improved or whether one class traded precision for recall. Classwise evidence is available for the separately trained SDI-4 + EXT-3-Original experiment in Section 8. That evidence helps describe plausible BG/WSSV/WSSV_BG confusions, but it cannot substitute for the missing classwise report of the primary experiment because the dataset composition and test partition differ.

4.3 Confusion-Matrix Analysis

No verified count or normalized confusion matrix was retained for the final 173-image primary checkpoint. Accordingly, this chapter does not fabricate a matrix from overall metrics. The retained normalized matrix belongs to the SDI-4 + EXT-3-Original extension and is analyzed in Section 8. A primary matrix should be regenerated only from the frozen prediction file and class order, not estimated from rounded per-class metrics.

4.4 Model Complexity and Efficiency

ASL-LDAM is training-only and adds no inference-time operators. SimAM is parameter-free but not computation-free because it computes feature statistics, a sigmoid map, and element-wise products. The DCFR texture path and channel gate contain trainable convolutions. A matched end-to-end profile of the final CE and final SimAM-DCFR checkpoints under one timing script was not retained, so an exact proposed-overhead delta is not reported.

The latency and FPS values in Table 6.3 are retained auxiliary GPU screening measurements for backbone checkpoints under batch-one inference. The final training and reproducibility narrative uses the accepted T4x2 / dual NVIDIA T4 protocol where applicable, but no unsupported T4x2 inference timing is fabricated here. The Android observation in Section 14 measures a different pipeline that includes device execution and application overhead; neither timing result should be transferred across hardware or tasks.

5. Classification Component Analysis

The retained component records were not all produced under a proven one-factor design, and the final selected checkpoint was distinct from the historical component-run series. Table 6.5 therefore uses the term "recorded component configurations" rather than "strict ablation." The obsolete full-combination component value is excluded according to the final evidence policy.

Table 6.5: Verified classification component configurations

Configuration	ASL-LDAM	SimAM	DCFR	Macro-F1 (%)	Interpretation boundary
CE reference	No	No	No	89.02	Controlled reference objective
ASL-LDAM	Yes	No	No	89.91	Loss-side configuration
ASL-LDAM + SimAM	Yes	Yes	No	89.73	Loss plus parameter-free attention
ASL-LDAM + DCFR	Yes	No	Yes	86.76	Loss plus trainable texture recalibration
Final ASL-LDAM + SimAM-DCFR	Yes	Yes	Yes	**91.01**	Separately selected final checkpoint; not a matched component-run row

ASL-LDAM alone was 0.89 points above CE in its retained run, while ASL-LDAM + SimAM was 0.71 points above CE. The ASL-LDAM + DCFR configuration was 2.26 points below CE, showing that the texture branch was not independently reliable in that recorded setting. The final selected checkpoint exceeded CE by 1.99 points, but the different provenance prevents attributing that gain uniquely to one component. The evidence supports the final integrated configuration under its recorded protocol; it does not establish an additive contribution model.

6. Classification Error Analysis

Three evidence constraints shape the error analysis. First, the primary prediction file was not retained, so exact error transitions are unavailable. Second, the background-removal process can suppress irrelevant context but can also remove weak edge or texture information. Third, BG, WSSV, and WSSV_BG share visible patterns, and a single image may contain small lesions, pose variation, occlusion, or illumination effects that make the class boundary ambiguous.

The SDI-4 + EXT-3-Original classwise report in Section 8 shows high WSSV precision but lower

WSSV recall, together with high WSSV_BG recall and lower precision. That pattern is consistent with some single-disease images being absorbed by the co-infection class, or with mixed visual evidence being over-predicted. It is a dataset-specific observation rather than proof that the same transition dominated the original 173-image test set.

Qualitative XAI panels indicate that the classifier often responds to shrimp-body regions rather than the entire background. However, some methods produce broader activation than others, and a highlighted region can reflect correlation, image contrast, or model sensitivity rather than a disease mechanism. Future prediction-level failure analysis should combine exact transitions, confidence, original and background-removed images, and multiple explanation methods once the required artifacts are available.

7. Classification Robustness under Controlled Synthetic Corruptions

The clean test images were transformed by five controlled corruption families while labels were preserved. These experiments measure sensitivity to selected synthetic degradations; they are not field validation and do not cover compression, color cast, turbidity, sensor differences, or every farm background.

Table 6.6: Classification robustness under selected synthetic corruptions

Corruption	**CE Macro-F1**	**Proposed Macro-F1**	**Δ (pp)**	**CE Kappa** $\times$100	**Proposed Kappa** $\times$100
Impulse noise	48.50	**55.82**	+7.32	41.96	**48.44**
Gaussian noise	74.79	**78.50**	+3.71	67.75	**71.82**
Contrast reduction	82.07	**88.10**	+6.03	76.90	**84.68**
Defocus blur	83.00	**88.66**	+5.66	78.02	**85.33**
Low light	83.15	**90.22**	+7.07	77.59	**87.11**

The proposed classifier was higher for all five retained aggregate conditions. The largest displayed Macro-F1 differences occurred under impulse noise (+7.32 points), low light (+7.07), and contrast reduction (+6.03). Impulse noise remained difficult for both models, with the proposed score still far below clean performance. A positive delta therefore does not imply that the corrupted condition was solved.

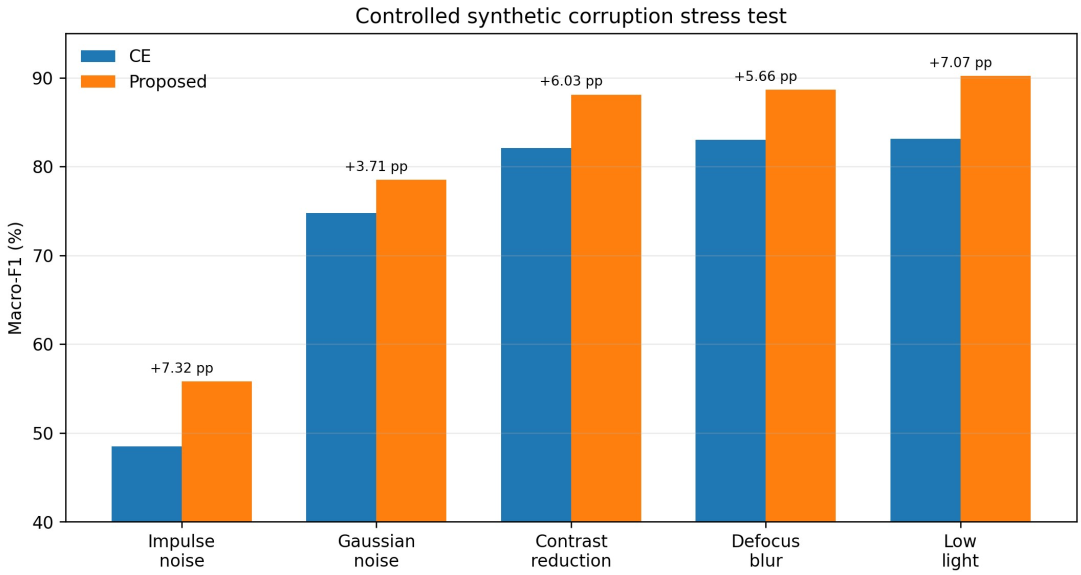

Figure 6.3: Baseline and proposed Macro-F1 under the five retained classification corruption families.

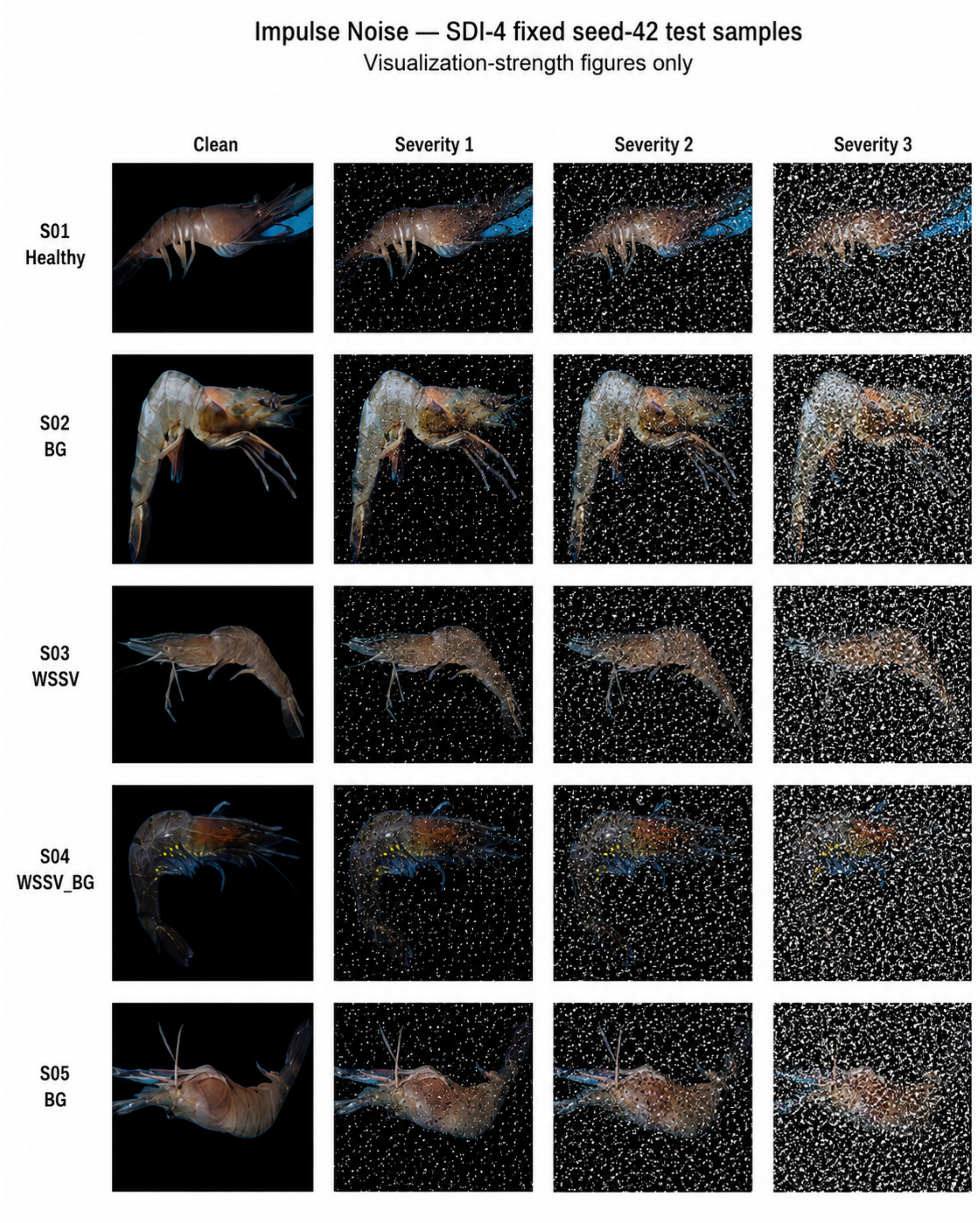

Figure 6.4: Visualization-strength impulse-noise examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

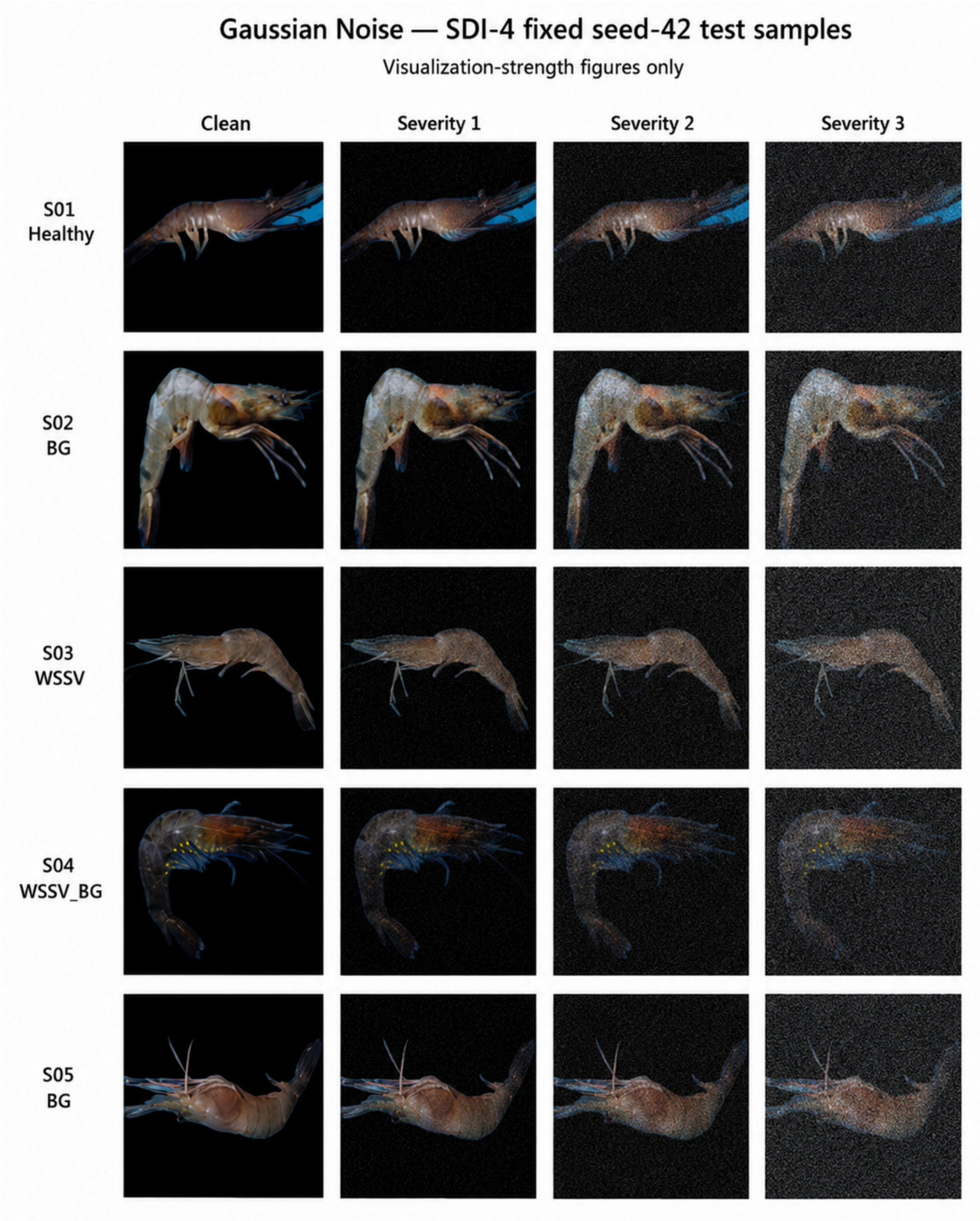

Figure 6.5: Visualization-strength Gaussian-noise examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

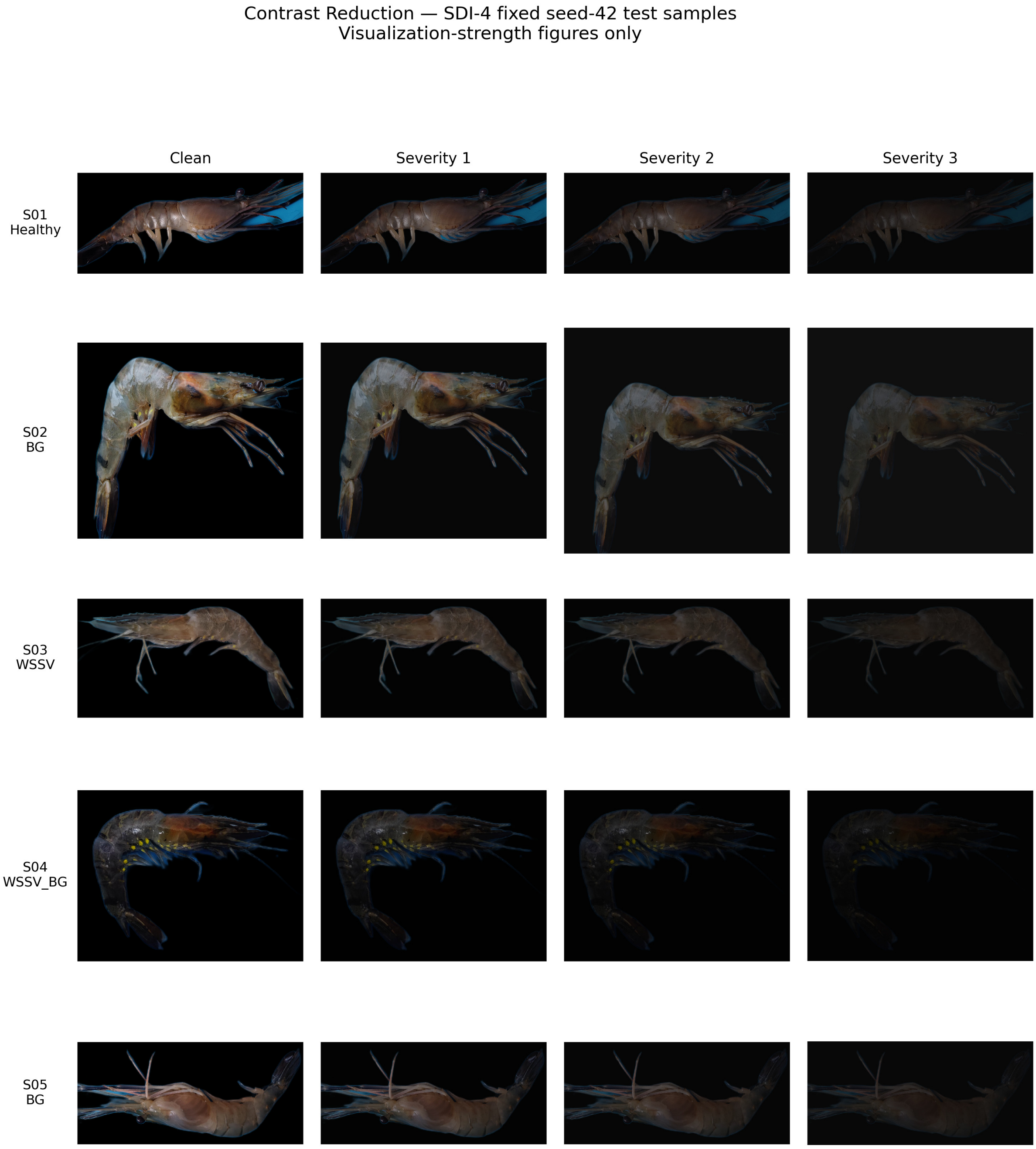

Figure 6.6: Visualization-strength contrast-reduction examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

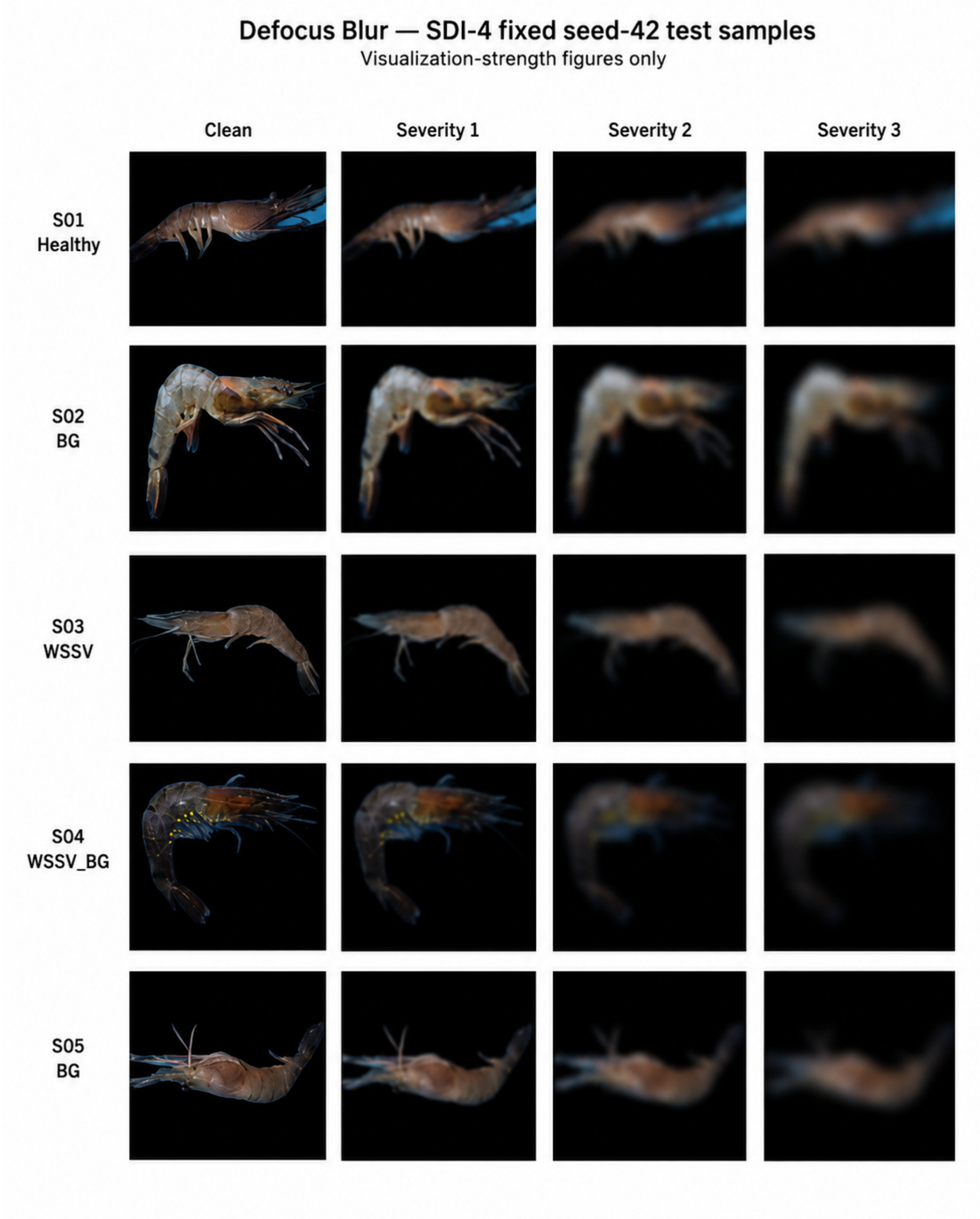

Figure 6.7: Visualization-strength defocus-blur examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

Figure 6.8: Visualization-strength low-light examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

8. Additional-Source and Partition-Wise Integrated Classification Evaluation

Table 6.7: Classification results on EXT-3-Original and SDI-4 + EXT-3-Original

Dataset regime	Method	Accuracy (%)	Macro-F1 (%)	Test images
EXT-3-Original	YOLO26m-cls + CE	70.21	72.30	47
EXT-3-Original	YOLO26m-cls + ASL-LDAM + SimAM-DCFR	**91.49**	**91.29**	47
SDI-4 + EXT-3-Original	YOLO26m-cls + CE	**87.73**	**86.51**	220
SDI-4 + EXT-3-Original	YOLO26m-cls + ASL-LDAM + SimAM-DCFR	83.18	82.18	220

The opposite outcomes in the two additional-source regimes require a source-aware interpretation rather than the shorthand explanation that one dataset is simply "easier." EXT-3-Original is a three-class retraining problem without WSSV_BG, whereas the integrated regime restores the four-class task and changes the source mixture, class priors, acquisition backgrounds, and the proportion of each class contributed by each source. WSSV_BG remains supplied only by SDI-4. ASL-LDAM uses class-dependent margins and the final classifier also changes feature recalibration; both mechanisms can interact with a changed empirical class distribution after retraining. However, no matched classwise causal experiment was retained that would isolate one of these factors as the unique cause of the 4.33-point integrated-regime deficit.

The YOLO26m-cls versus YOLO26x-cls train/validation diagnostic in Figure 6.1A provides an additional capacity-related boundary. A substantially larger classifier did not improve validation performance under the same fixed protocol, so the integrated-regime drop should not be attributed to insufficient parameter count alone. The evidence instead supports a more conservative conclusion: performance depends on the joint dataset regime, class composition, optimization objective, and source characteristics, and the proposed configuration is not uniformly superior across retrained source combinations.

On EXT-3-Original, the proposed configuration exceeded CE by 18.99 percentage points in Macro-F1. On SDI-4 + EXT-3-Original, CE exceeded the proposed configuration by 4.33 points. This reversal is central to the interpretation: the integrated method improved the primary SDI-4 clean result and the separately trained three-class external experiment, but it did not dominate the partition-wise integrated four-class regime.

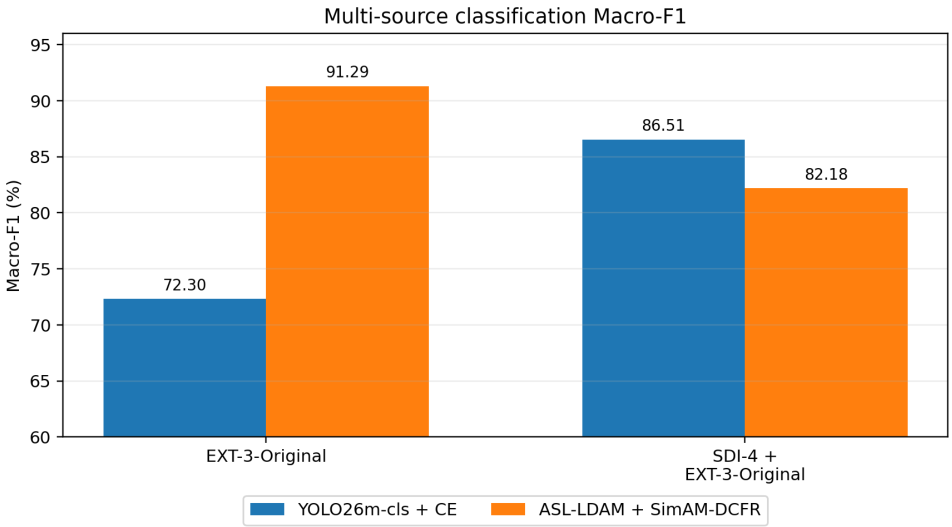

Figure 6.9: Macro-F1 of CE and the proposed configuration on EXT-3-Original and SDI-4 + EXT-3-Original.

The 47-image EXT-3-Original test partition is sensitive to a small number of prediction changes. The 220-image integrated test partition is larger but changes source composition and class priors. Neither experiment is direct external inference by the original SDI-4 checkpoint.

Table 6.8: Classwise performance of the proposed classifier on SDI-4 + EXT-3-Original

Class	Precision (%)	Recall (%)	F1-score (%)	Support
Healthy	88.00	91.67	**89.80**	72
BG	80.85	84.44	82.61	45
WSSV	**90.91**	71.43	80.00	70
WSSV_BG	67.44	87.88	76.32	33
Macro average	81.80	83.85	82.18	220

The classwise report shows that WSSV achieved high precision but lower recall, while WSSV_BG achieved comparatively high recall with lower precision. This pattern is consistent with confusion between disease-specific and co-infection evidence. The table belongs only to the SDI-4 + EXT-3-Original experiment and does not replace the missing classwise ledger for the primary 173-image SDI-4 test partition.

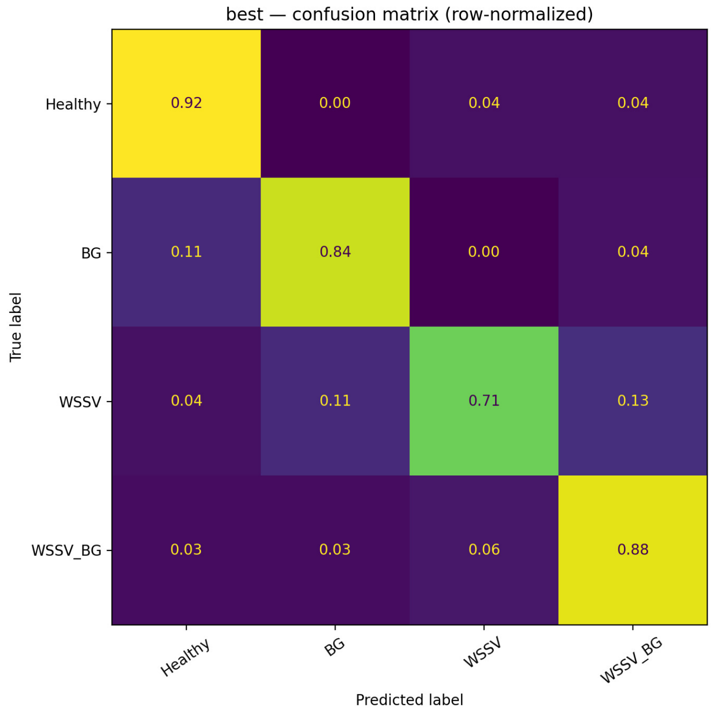

Figure 6.10: Row-normalized confusion matrix for the proposed classifier trained and evaluated on SDI-4 + EXT-3-Original.

The matrix must not be presented as the primary SDI-4 confusion matrix. It is retained as supporting evidence for the partition-wise integrated regime and demonstrates that aggregate accuracy can conceal class-specific precision–recall trade-offs.

9. Explainability Analysis

Grad-CAM++ [86], HiResCAM [96], and EigenCAM [88] were generated for correctly classified SDI-4 + EXT-3-Original samples. The methods answer related but different questions: gradient-weighted class activation, spatially detailed activation-gradient products, and class-independent principal activation structure. Figure 6.11 places the methods on common samples so that agreement and disagreement can be inspected directly.

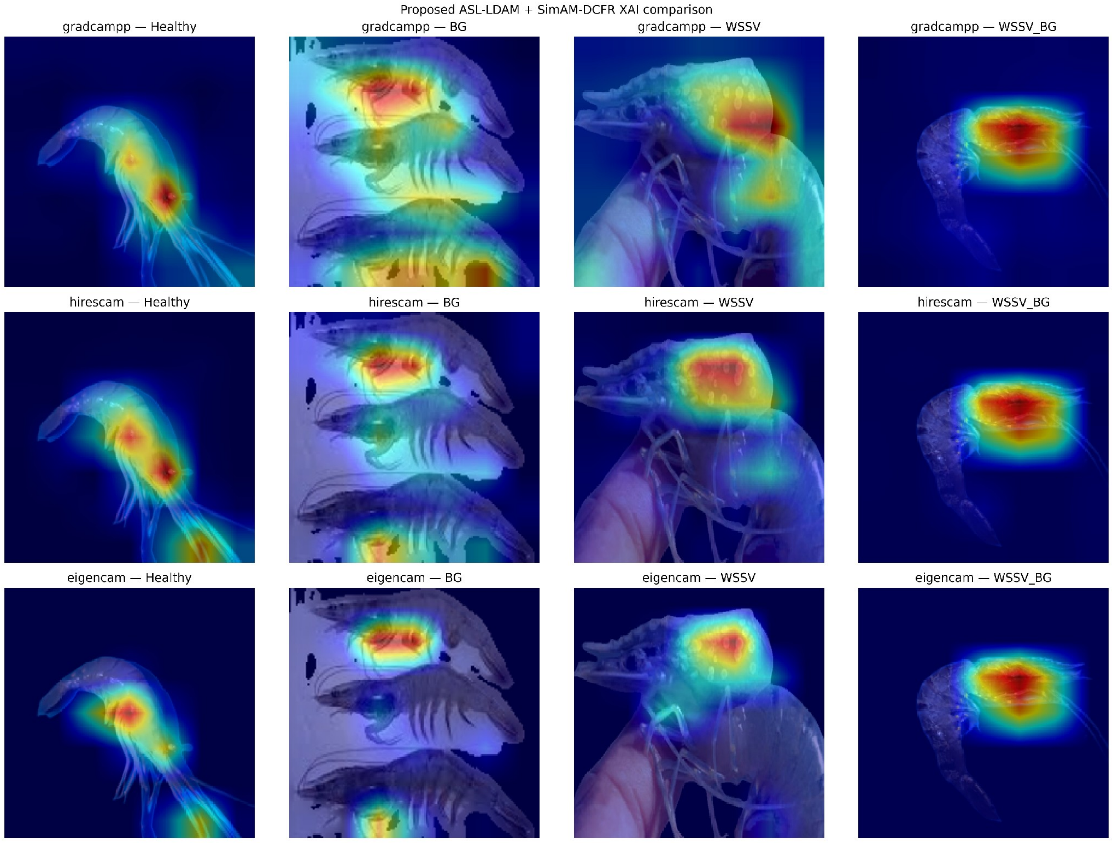

Figure 6.11: Cross-method qualitative comparison of Grad-CAM++, HiResCAM, and EigenCAM on retained classification samples.

Agreement over shrimp-body regions is useful as a sanity check against exclusive background reliance. Disagreement in spatial extent is expected because the methods aggregate features differently. Broad heatmaps should not be interpreted as segmentation masks, and concentrated activation does not establish causal reliance on a clinically meaningful symptom. The XAI evidence is qualitative, was generated on selected samples, and does not estimate localization accuracy.

10. Instance-Segmentation Baselines and Leakage Analysis

10.1 Segmentation Baseline Comparison

Six Ultralytics segmentation candidates were evaluated under the grouped split. Table 6.9 combines full-test and labeled-only localization with the Healthy false-positive rate and the project-defined HScore. HScore is an internal decision criterion, not an external benchmark.

Table 6.9: Segmentation baseline comparison under the grouped split

Model	Params (M)	Full mAP50	Labeled mAP50	mAP50-95	Healthy FP (%)	HScore (%)
YOLO11n-seg	2.877	**48.2**	**51.2**	16.8	24.4	**46.0**
YOLO26s-seg	11.506	41.7	46.0	**18.2**	26.8	39.7
YOLOv8n-seg	3.410	39.5	43.2	13.9	34.1	35.3
YOLOv8s-seg	11.821	39.1	44.4	15.9	34.1	36.4
YOLO26n-seg	3.126	43.0	46.3	16.0	**19.5**	39.3
YOLO11s-seg	10.113	29.7	36.1	12.4	51.2	25.0

Table 6.9B. Additional instance-segmentation comparators on the grouped leakage-safe split

Model	Params (M)	GFLOPs	Full mAP50	Labeled mAP50	Labeled mAP50-95	Recall	Healthy FP (%)
RTMDet-Ins-Tiny	5.615	11.8	44.90	50.90	16.40	64.71[a]	95.12
RTMDet-Ins-S	10.153	21.5	48.30	54.90	18.70	64.71[a]	100.00
YOLO26n-seg	3.054	10.3	35.43	36.88	13.00	36.00[b]	9.76
YOLO26s-seg	11.435	37.3	40.92	44.10	16.70	48.40[b]	24.39

[a] RTMDet recall was computed at IoU 0.50 and confidence 0.25. [b] YOLO mask recall is the Ultralytics-exported recall. These recall values are reported for context but are not treated as numerically identical because the evaluation implementations differ. MMEngine evaluator time for RTMDet is not pure model-inference latency and is therefore excluded from the speed comparison.

RTMDet-Ins-S produced the strongest localization result among the four additional comparators, reaching 48.30% full-test mAP50, 54.90% labeled mAP50, and 18.70% labeled mAP50-95. This gain was accompanied by a 100.00% Healthy false-positive rate in the retained Healthy-negative diagnostic, meaning that every Healthy negative image produced at least one retained disease prediction under that evaluation. RTMDet-Ins-Tiny showed the same trade-off direction, with 95.12% Healthy FP. In contrast, the original YOLO11n-seg reference in Table 6.9 achieved 48.2% full mAP50 and 51.2% labeled mAP50 with a substantially lower 24.4% Healthy FP rate and only 2.877M parameters. YOLO11n-seg was therefore retained as the attention reference because localization quality, model compactness, and Healthy-negative behavior were more balanced; the selection does not imply dominance on every individual metric.

Because the additional comparator set uses evaluation implementations with non-identical recall definitions, Table 6.9B is interpreted as a separate comparator study rather than a direct extension of the Ultralytics rank order in Table 6.9. The architecture decision is consequently multi-objective: localization performance is evaluated jointly with parameter efficiency and Healthy-negative robustness. RTMDet-Ins-S optimization and validation-based checkpoint selection are examined in Section 10.1.1, and the final architecture trade-off is analyzed in Section 10.1.2.

10.1.1 RTMDet-Ins-S Training and Validation Evidence

RTMDet-Ins-S was trained and evaluated on the same specimen-grouped leakage-safe partition used for the additional comparator study: 331 training groups (905 images, 810 instances), 40 validation groups (115 images, 102 instances), and 45 test groups (129 images, 119 instances). The test partition contained 88 disease-labeled images and 41 Healthy empty-label negatives. The model used 640 x 640 inputs, 100 configured epochs, physical batch size 4, and gradient accumulation of 4, corresponding to an effective batch size of 16. The two predicted instance classes were BG and WSSV.

Figure 6.11A. RTMDet-Ins-S training-loss trajectories over 100 epochs. The total objective and its classification, bounding-box, and mask components are reported from the retained MMDetection log.

The training curves show rapid optimization during the initial epochs followed by a slower decline. The retained log decreases from approximately 3.08 total loss at the beginning of training to approximately 1.23 near epoch 100. Classification loss falls most sharply during the early phase, while bounding-box and mask losses decline more gradually. This behavior is consistent with convergence of the training objective, but it is not by itself evidence of better held-out segmentation quality. The historical MMDetection run used for Table 6.9B did not record validation loss; checkpoint selection therefore relied on validation COCO mask mAP50 rather than on a train-versus-validation loss minimum.

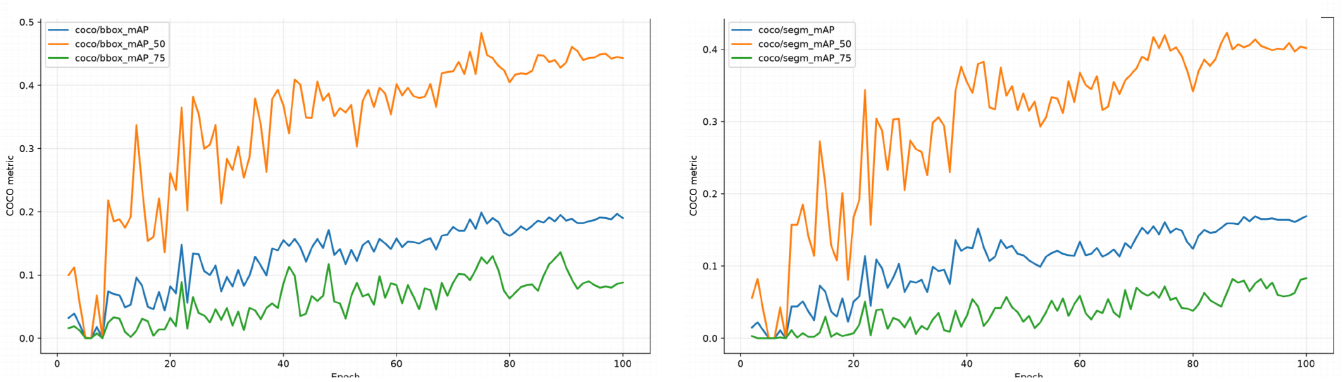

Figure 6.11B. RTMDet-Ins-S validation COCO trajectories. Left: bounding-box mAP, mAP50, and mAP75. Right: mask mAP, mAP50, and mAP75. These curves are validation AP metrics, not validation-loss curves.

Validation performance was substantially noisier than the training objective during the early epochs and then stabilized. The retained validation mask mAP50 reached its maximum of 42.30% at epoch 86. At that checkpoint, mask mAP50-95 was 15.90% and mask mAP75 was 6.30%; the corresponding bounding-box metrics were 44.70% mAP50, 18.30% mAP50-95, and 9.90% mAP75. By epoch 100, validation mask mAP50 had decreased to 40.20%. Consequently, **best_coco_segm_mAP_50_epoch_86.pth**, rather than the final epoch, was used to obtain the RTMDet-Ins-S test results reported in Table 6.9B. This distinction prevents the endpoint test row from being interpreted as an arbitrary epoch-100 result.

10.1.2 Selection of YOLO11n-seg over RTMDet-Ins-S

RTMDet-Ins-S is a strong localization comparator, but model selection in this study is multi-objective rather than mAP-only. Relative to the original YOLO11n-seg grouped baseline, RTMDet-Ins-S improves full-test mAP50 only marginally (48.30% versus 48.2%) while increasing labeled mAP50 from 51.2% to 54.90% and labeled mAP50-95 from 16.8% to 18.70%. The principal failure is Healthy-negative robustness: RTMDet-Ins-S produced at least one retained disease prediction on all 41 Healthy test images, corresponding to a 100% Healthy false-positive rate, whereas YOLO11n-seg recorded 24.4%. The comparator is also substantially larger, with 10.153M parameters compared with 2.877M for YOLO11n-seg. Thus, the higher disease-image localization score did not compensate for the 75.6-percentage-point deterioration in Healthy false-positive rate and the markedly larger model footprint.

The later CA -> SimAM refinement further strengthens this engineering choice: the selected YOLO11n-seg + CA -> SimAM model uses 2.855M parameters and approximately 9.7 GFLOPs while reaching 55.07% labeled mAP50 and 19.64% labeled mAP50-95 (Table 6.21), both slightly above the RTMDet-Ins-S labeled metrics. Because framework implementations and recall definitions are not perfectly identical, this comparison is not presented as a universal architecture ranking. Within the retained CVio evaluation, however, YOLO11n-seg provides the more appropriate foundation for a lightweight mobile-oriented system because it combines competitive localization, substantially safer Healthy-negative behavior, and much lower complexity.

YOLO11n-seg had the highest full and labeled mAP50 and HScore in this baseline table. YOLO26s-seg had the highest mAP50-95, while YOLO26n-seg had the lowest Healthy false-positive rate. Thus, the baseline screen did not yield one model that dominated every criterion. YOLO11n-seg was retained as the attention reference because its overall localization-efficiency trade-off was strongest within this recorded set.

10.2 Split-Policy Leakage Comparison

Table 6.10: Leakage comparison across segmentation split policies

Split policy	Train-test overlap (%)	Full mAP50	Labeled mAP50	mAP50-95	Healthy FP (%)
Plain random image	98.7 ± 1.2	67.1 ± 1.3	67.7 ± 1.3	29.9 ± 2.4	5.4 ± 2.7
Stratified random image	94.0 ± 6.1	65.1 ± 6.0	65.9 ± 6.2	28.2 ± 1.6	9.8 ± 2.4
Stratified grouped specimen	**0.0 ± 0.0**	45.1 ± 2.7	47.3 ± 3.5	14.4 ± 2.1	23.6 ± 11.0

Plain random splitting exceeded grouped splitting by 20.4 points in labeled-only mAP50 while retaining almost complete train-test specimen overlap. Bold in the overlap column marks the validity objective rather than a performance ranking. The lower grouped result is not evidence that grouping damages the model; instead, it indicates that repeated-view leakage made the random-split problem easier. The grouped protocol is the principal evaluation because it better approximates performance on previously unseen filename-derived specimens.

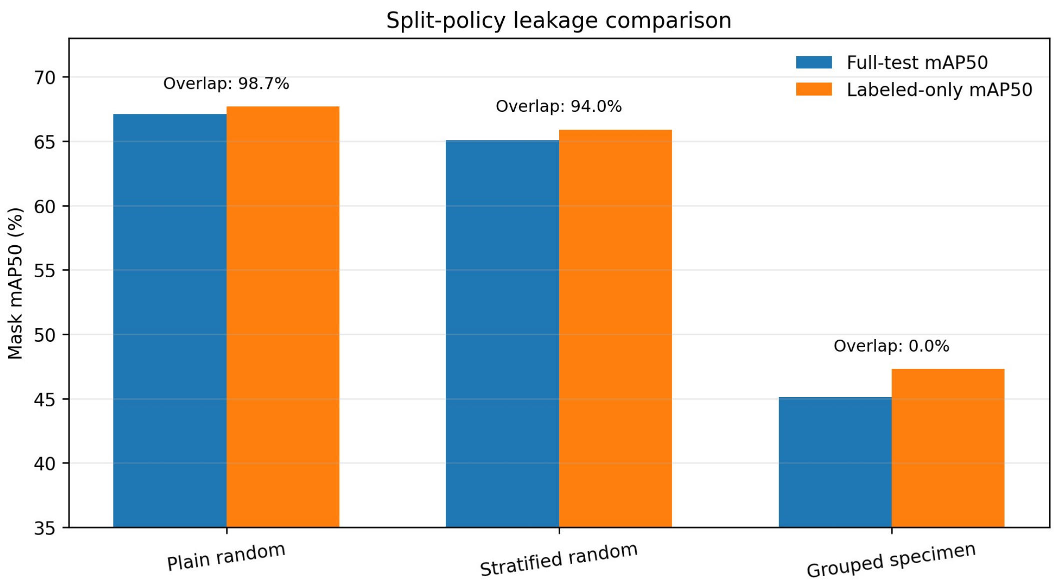

Figure 6.12: Full-test and labeled-only mask mAP50 under image-level and grouped-specimen split policies.

The grouped split also produced a higher and more variable Healthy false-positive rate, exposing a failure mode hidden by the leaky splits. This supports reporting negative-image diagnostics together with mAP rather than selecting a split solely because it yields larger scores.

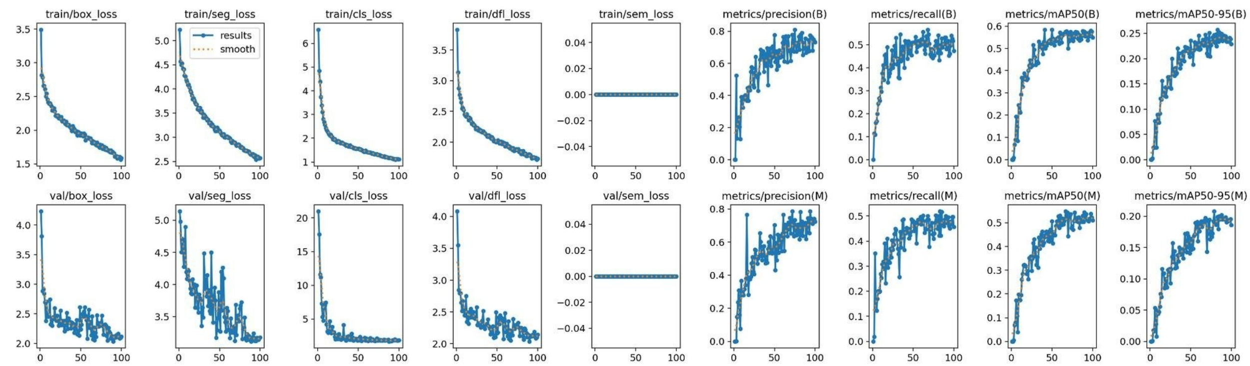

Figure 6.13: Representative training and validation curves retained for convergence inspection.

Figure 6.13 is a diagnostic artifact rather than a held-out result. Smooth convergence does not correct data leakage, annotation bias, or a weak test protocol.

11. Attention-Based Instance-Segmentation Results

The attention runs were not fully identical. They differed in augmentation, epoch budget, early stopping, software versions, module placement, and whether an auxiliary loss was introduced. Table 6.11 preserves these differences rather than presenting the study as a perfect one-factor ablation.

Table 6.11: Recorded attention configuration protocols

Configuration	Placement / objective	Aug. policy	Configured / executed epochs	Init. source	Ultralytics / PyTorch
YOLO11n-seg reference	No added attention	Clean-light	100 / 60, early stop	yolo11n-seg.pt	8.4.61 / 2.10.0
CA to SimAM	P3, P4, P5	Strong	100 / 100	594/594 tensors	8.4.67 / 2.11.0
LKA to SimAM	P3, P4, P5	Clean-light	100 / 75, early stop	579/579 tensors	8.4.61 / 2.10.0
DPCA	P3, P4, P5	Strong	200 / 200	yolo11n-seg.pt	8.4.102 / 2.10.0
CoTEGate + BoundaryLite	P4 + auxiliary loss	Strong	200 / 184, early stop	yolo11n-seg.pt	8.4.95 / 2.10.0

Table 6.12: Attention localization and efficiency summary on the original grouped dataset

Model	Full mAP50	Labeled mAP50	Labeled mAP50-95	Recall	Parameters	Inf. ms/image	FPS
YOLO11n-seg baseline	44.09	46.64	16.90	41.0	2,842,998	**6.0**	**166.7**
CA → SimAM	49.71	**55.07**	**19.64**	49.7	2,854,718	9.3	107.5
LKA → SimAM	**50.03**	52.49	18.12	**54.4**	2,963,510	8.6	116.3
DPCA	48.80	52.25	15.80	51.3	2,848,377	8.8	113.6
CoTEGate + BoundaryLite	49.21	51.74	16.60	49.7	**2,736,562**	**8.2**	**122.0**

The efficiency columns were added to answer the Council's request for parameter-level comparison. CA → SimAM achieved the highest labeled mAP50 (55.07%) and labeled mAP50-95 (19.64%), while increasing the reference parameter count by only 11,720 parameters. LKA → SimAM led full-test mAP50 and recall but had the largest parameter count in this attention set. CoTEGate + BoundaryLite used the fewest parameters and was the fastest attention variant in the retained notebook timing, whereas the unmodified baseline remained fastest overall. The recorded millisecond/FPS values are notebook GPU measurements and are not Android benchmarks. Because augmentation, training budget, initialization, software versions, placement, and auxiliary losses differ across rows, the table supports configuration-level comparison rather than a strict one-factor causal ablation.

Table 6.12B. Placement ablation for CA → SimAM on YOLO11n-seg

Placement	Full mAP50	Labeled mAP50	Labeled mAP50-95	Recall	Parameters	Inf. ms/image	FPS
P3/P4/P5	**49.71**	**55.07**	**19.64**	**49.7**	2,854,718	9.3	107.5
Neck concat	40.90	43.30	14.20	46.5	2,818,718	10.3	97.1
Head P3	40.90	43.40	14.60	41.5	**2,691,582**	9.9	101.0
Head P4	47.70	50.50	16.80	47.4	2,693,246	**9.6**	**104.2**
After C2PSA	43.80	45.80	16.90	45.6	2,696,574	10.3	97.1
Backbone P3	38.50	40.90	13.60	43.6	**2,691,582**	10.6	94.3
Neck P3/P4	44.30	47.30	15.90	43.6	2,721,250	10.4	96.2

The P3/P4/P5 placement led all four retained localization/recall columns in this placement study. Among single placements, Head P4 was the strongest localization choice, but it remained below the multi-level P3/P4/P5 configuration. The result supports the selected multi-scale placement within this controlled placement experiment; it does not convert the broader attention-family comparison into a strict ablation.

Table 6.13: Healthy-negative and mask-count diagnostics on the original grouped dataset

Model	Healthy FP images (%)	FP masks	Disease misses (%)	Full MAE	Labeled MAE	HScore
Reference	15 (36.59)	17	11 (12.50)	0.354167	0.354167	0.3934
CA to SimAM	14 (34.15)	16	12 (13.64)	0.333333	**0.306818**	**0.4808**
LKA to SimAM	15 (36.59)	18	**7 (7.95)**	0.475452	0.492424	0.4518
DPCA	**11 (26.83)**	**12**	12 (13.64)	**0.313953**	0.323864	0.4590
CoTEGate + BoundaryLite	15 (36.59)	17	10 (11.36)	0.395349	0.386364	0.4540

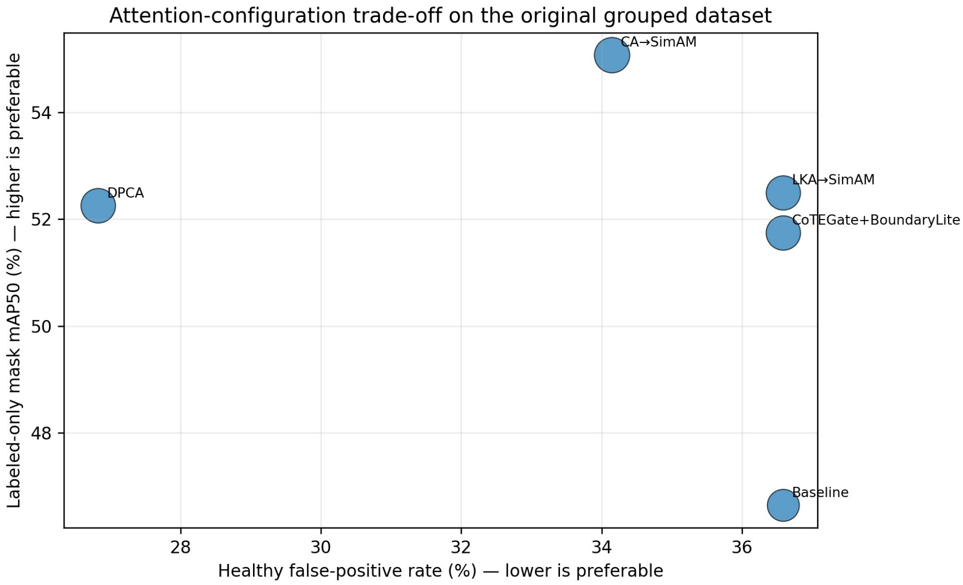

Figure 6.14: Trade-off between labeled-only mAP50, Healthy false-positive rate, and HScore for the original grouped attention runs.

CA-to-SimAM had the highest HScore, DPCA the lowest Healthy false-positive rate, and LKA-to-SimAM the lowest disease missed-image rate. No method dominated all diagnostic axes. CoTE-Gate + BoundaryLite changed both architecture and training objective, so its row is not an attention-only estimate.

Table 6.14: Attention results on the combined segmentation dataset

Model	Full mAP50	Labeled mAP50	Labeled mAP50-95	Healthy FP (%)	HScore
CA to SimAM	**62.99**	**65.94**	**25.72**	39.0	**0.5913**
LKA to SimAM	56.79	58.50	25.02	39.0	0.5110
DPCA	50.46	53.14	19.15	48.8	0.4471
CoTEGate + BoundaryLite	43.99	48.00	17.33	**36.6**	0.3966

The combined dataset contained 1,452 images and used a mixed grouped–stratified allocation because 303 additional diseased images did not follow the original filename convention. CA-to-

SimAM led the recorded localization metrics and HScore in this extension, but the changed dataset and split prevent a direct attention-only comparison with the original grouped table.

12. Fourier-Domain Instance-Segmentation Results

12.1 Fourier Candidate Screening

The strict no-augmentation, hook-off screen isolated image-space frequency preprocessing. Each model received the intended transformed evaluation input. Table 6.15 reports the complete retained candidate table.

Table 6.15: Fourier candidate screening under the strict control

Configuration	Labeled mAP50	mAP50-95	Healthy FP (%)	Disease miss (%)	Healthy-aware score (%)
No Fourier control	23.54	6.06	39.02	22.73	14.08
High-pass $\sigma = 50$, $\alpha = 0.05$	30.88	7.92	34.15	25.00	20.90
W1 high-pass $\sigma = 50$, $\alpha = 0.10$	**34.51**	8.85	63.41	**19.32**	**21.08**
High-frequency damping	32.11	**10.68**	39.02	52.27	17.40
Band-pass 12/60, $\alpha = 0.20$	29.52	8.39	26.83	43.18	17.21
Low-frequency flattening	27.87	8.52	**14.63**	46.59	16.66
Homomorphic filtering	26.12	9.16	41.46	23.86	15.06
Gaussian low-pass $\sigma = 25$	25.08	8.56	31.71	37.50	15.37

W1 had the highest labeled mAP50 and healthy-aware score in this isolated screen, improving mAP50 by 10.97 points. However, its Healthy false-positive rate increased to 63.41%, which is operationally severe. High-frequency damping achieved the highest mAP50-95 but missed more than half of the diseased images. The screen therefore selected W1 as a high-pass reference for interaction studies, not as a universally superior preprocessing method.

12.2 W1 High-Pass Reference

W1 used a Gaussian low-pass mask with $\sigma = 50$ and residual enhancement $\alpha = 0.10$. Because the operation changed pixel intensities without geometric warping, polygon coordinates remained aligned. Its strongest isolated benefit occurred when conventional augmentation and the hidden Albumentations hook were both disabled, suggesting that W1 supplied a perturbation absent from the strict control. The accompanying increase in false-positive masks indicates that enhanced high-frequency detail can amplify disease-like structures in Healthy images. This is the central W1 trade-off: greater sensitivity did not automatically produce safer negative-image behavior.

12.3 Fourier–Augmentation Order Study

Table 6.16: Interaction between W1 and augmentation regimes

Training regime	No Fourier mAP50	W1 mAP50	Δ (pp)	Healthy FP: off to W1	Healthy-aware: off to W1
No augmentation, hook off	23.54	**34.51**	+10.97	39.02 to 63.41	14.08 to 21.08
No augmentation, hook on	**30.61**	29.94	-0.67	36.59 to 41.46	19.73 to 17.53
Clean-light, hook off	**47.34**	40.82	-6.52	39.02 to 36.59	39.48 to 33.49
Clean-light, hook on	51.20	**53.18**	+1.98	24.39 to 24.39	45.96 to 47.02
Strong, hook off	**59.51**	57.71	-1.80	34.15 to 31.71	53.26 to 50.15

The effect of W1 depended on conventional augmentation. It improved the strict control and the clean-light hook-on condition, but reduced mAP50 under the clean-light hook-off and strong policies. Strong augmentation without W1 produced the highest labeled mAP50 in this table. This interaction rejects a simple claim that Fourier preprocessing consistently adds to spatial and photometric augmentation.

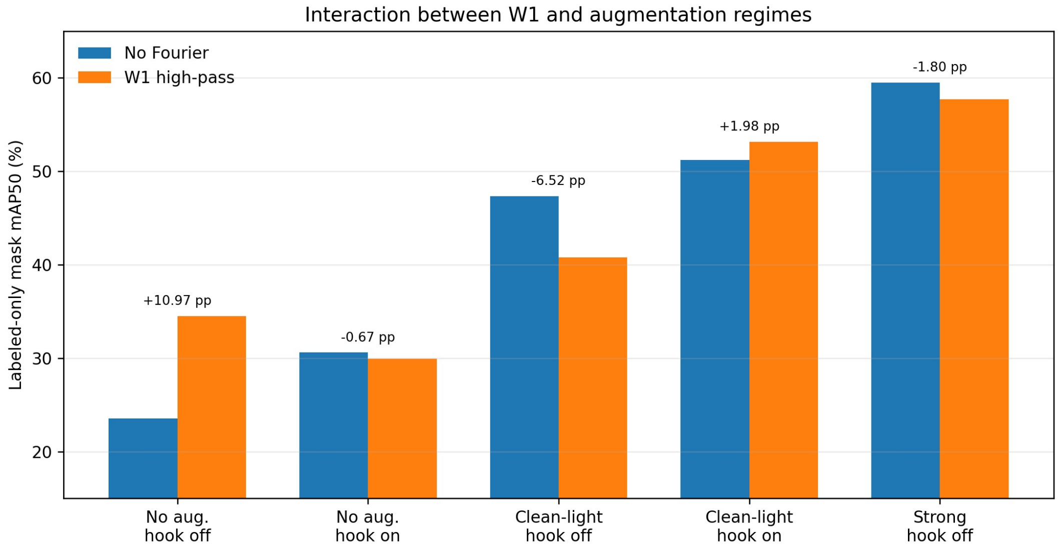

Figure 6.15: Labeled-only mask mAP50 with and without W1 across five augmentation and hook regimes.

The better order depended on augmentation type. W1 before spatial augmentation was favorable for FlipLR, rotation, and erasing, whereas W1 after augmentation was favorable for FlipUD. Translate and scale remained worse than their augmentation-only references.

Table 6.17: **Offline augmentation–Fourier order study**

N=1 augmentation	**Augmentation-only mAP50**	**Higher Fourier order**	**Fourier mAP50**	**Δ (pp)**
Translate 0.08	**37.82**	Augmentation to W1	35.00	-2.82
Scale 0.10	**36.91**	Augmentation to W1	31.85	-5.05
FlipLR 0.25	32.75	W1 to augmentation	**36.51**	+3.77
Rotation 3 degrees	31.08	W1 to augmentation	**35.95**	+4.87
Erasing 0.15	19.81	W1 to augmentation	**29.07**	+9.26
FlipUD 0.30	33.45	Augmentation to W1	**34.51**	+1.05

A large relative gain from a weak baseline, such as erasing 0.15, did not make that configuration the strongest overall model.

Direct Spectrum Analysis of the W1–Augmentation Interaction

The Council specifically asked why W1 improved the no-augmentation control but became slightly harmful under the strong-augmentation policy. The revised analysis therefore uses direct spatial- and frequency-domain evidence rather than inferring the mechanism from mAP alone. The retained paired study used 24 test images (six each from Healthy, BG, WSSV, and WSSV_BG) and 12 matched stochastic augmentation draws per image, yielding 288 paired observations. W1 was evaluated both alone and in combination with the recorded strong augmentation pipeline.

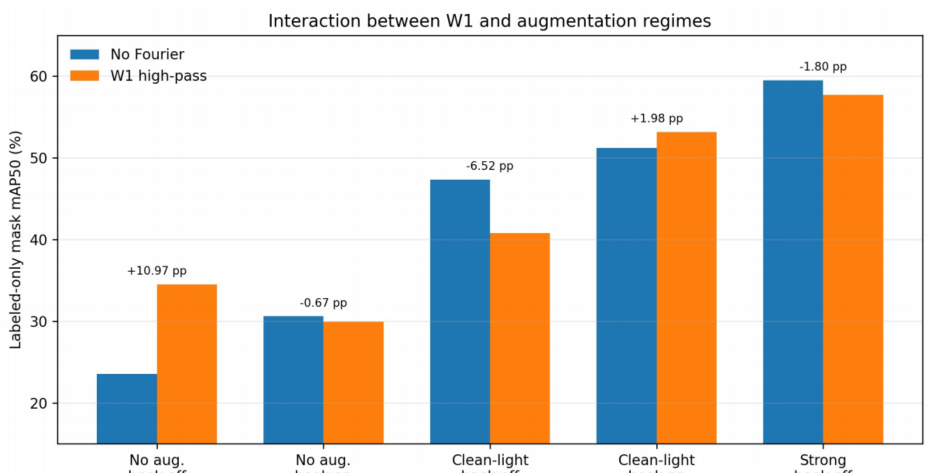

Figure 6.15A. Labeled-only mask mAP50 with and without W1 across the retained augmentation/hook regimes. W1 improved the strict no-augmentation, hook-off control from 23.54% to 34.51% (+10.97 pp), but reduced the strong-augmentation, hook-off result from 59.51% to 57.71% (-1.80 pp).

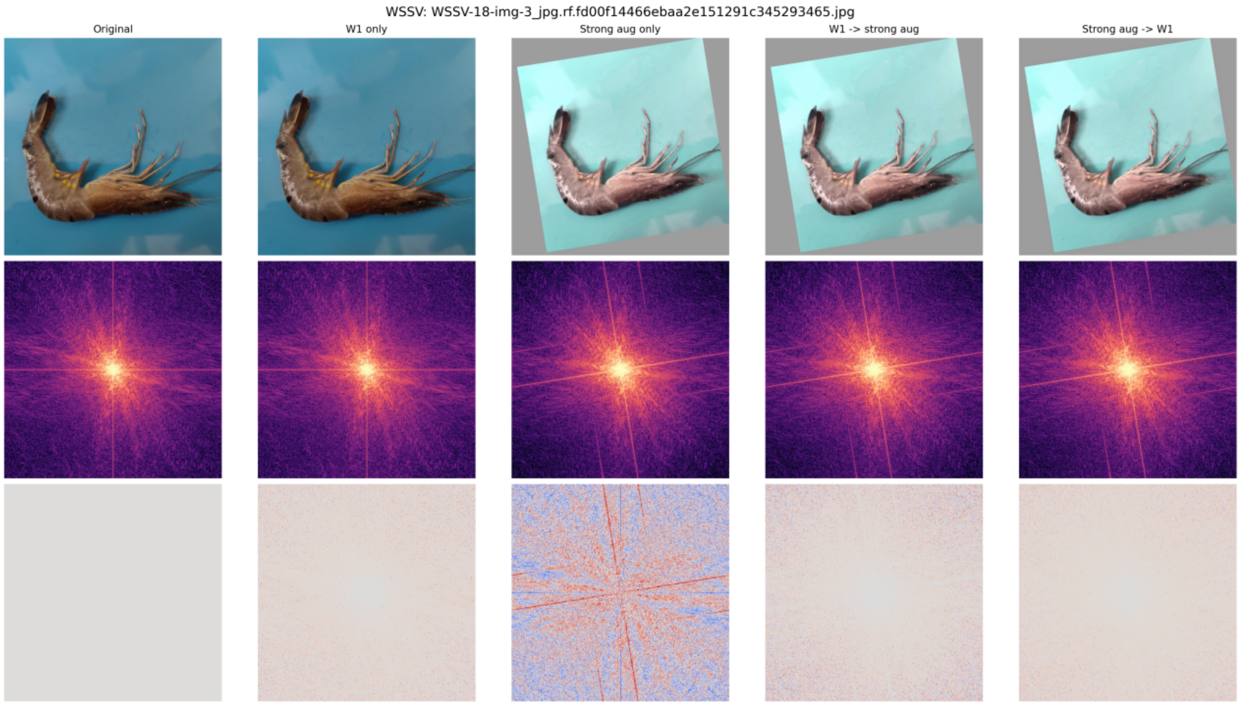

Figure 6.15B. Representative spatial and Fourier-domain comparison of Original, W1 only, Strong augmentation only, W1 followed by strong augmentation, and strong augmentation followed by W1. The log-power spectra and signed spectral differences show that the augmentation pipeline already redistributes substantial frequency content, while W1 adds a smaller residual emphasis when combined with the stronger transform sequence.

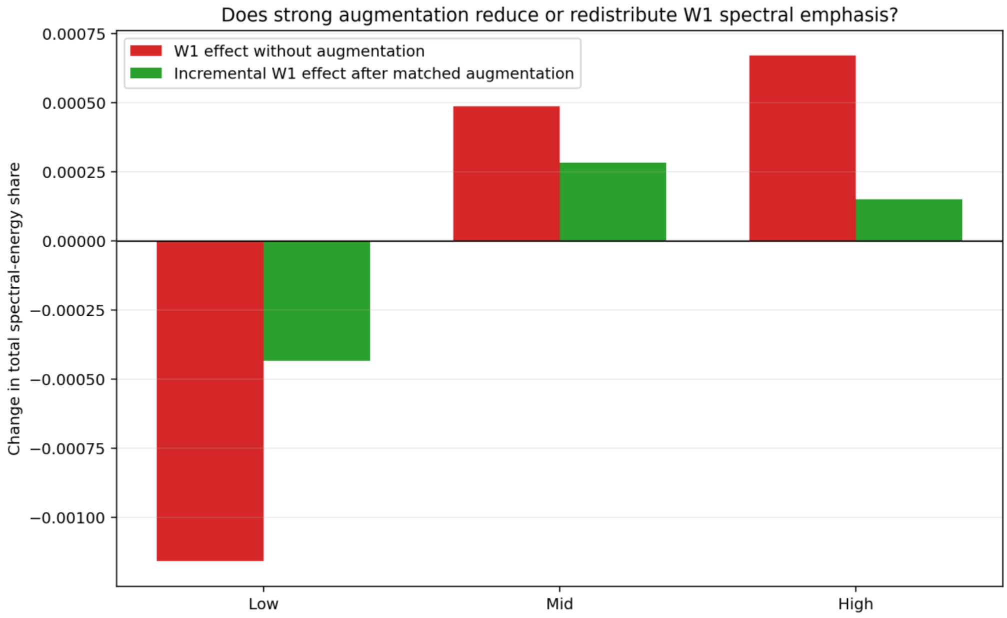

Figure 6.15C. Mean change in low-, mid-, and high-frequency energy share caused by W1 alone versus the incremental W1 effect after matched strong augmentation. The high-band increment after augmentation is much smaller than the W1-only increment.

Across the 288 paired observations, W1 alone increased the high-band energy share by an average of 0.000670, whereas the incremental increase attributable to W1 after strong augmentation was 0.000151, only 22.5% of the original W1 high-band effect. The retained edge-sensitive summaries showed the same attenuation pattern: the incremental Laplacian-variance effect was 21.1% of the W1-

alone effect, mean-gradient effect was 56.1%, and lesion-region gradient effect was approximately 79.0%. These measurements indicate that strong spatial/color augmentation does not completely eliminate W1, but it overlaps with or redistributes much of W1's global high-frequency emphasis. In the recorded pipeline, geometric transforms can resample already enhanced edges and HSV operations alter channel-intensity relationships. This explanation is consistent with the observed spectrum and seed-42 mAP interaction; it is not a multi-seed causal proof.

12.4 Feature-Domain Fourier Refinement

The feature-domain FDDEM direction was implemented separately from W1 input preprocessing. The retained implementation description specifies learnable complex gains in radial bands and residual feature reconstruction, but it does not contain a final, fully audited result table tied to one frozen checkpoint. No numerical FDDEM claim is therefore inserted. This omission preserves the distinction between a described architectural experiment and a verified selected outcome.

Table 6.18: Recorded Fourier and augmentation inference timing

Method	Fourier preprocessing (ms/image)	YOLO prediction (ms/image)	Effective total (ms/image)
W1 high-pass	48.62	38.77	87.38
Erasing 0.10	0.00	12.69	**12.69**
FlipUD 0.30	0.00	12.69	**12.69**
Scale 0.10 + Erasing 0.10	0.00	12.82	12.82

The timing protocol used the same 129-image grouped test set, CUDA execution, 640-pixel inputs, ten warm-up images, and 119 timed images. W1 added 48.62 ms of preprocessing and increased effective latency to 87.38 ms per image in that segmentation timing scope. This measurement is not interchangeable with the auxiliary classification screening latency or the Android runtime because the task, model, hardware context, and measured pipeline differ.

13. Segmentation Robustness and Failure Analysis

The segmentation robustness set differed from the classification corruption set. The retained positive-gain table contains color cast, turbidity, JPEG artifacts, overexposure, and Gaussian noise. Because the rows were selected by positive improvement, they cannot establish that the proposed segmentation configuration improved every evaluated corruption.

Table 6.19: Selected attention-based segmentation corruption results

Corruption	Reference labeled mAP50 (%)	Attention-based proposed labeled mAP50 (%)	Improvement (pp)
Color cast	29.81	**54.65**	**+24.84**
Turbidity	25.03	**44.97**	+19.94
JPEG artifact	36.58	**56.39**	+19.81
Overexposure	35.21	**52.46**	+17.25
Gaussian noise	18.13	**33.44**	+15.31

Table 6.19 belongs to the attention-based segmentation branch. Color cast produced the largest selected difference, followed by turbidity and JPEG artifacts. Gaussian noise remained difficult even after improvement. The table reports selected positive-gain conditions rather than a complete corruption benchmark; Fourier preprocessing is evaluated separately in Section 12 and is not merged into the attention-based column.

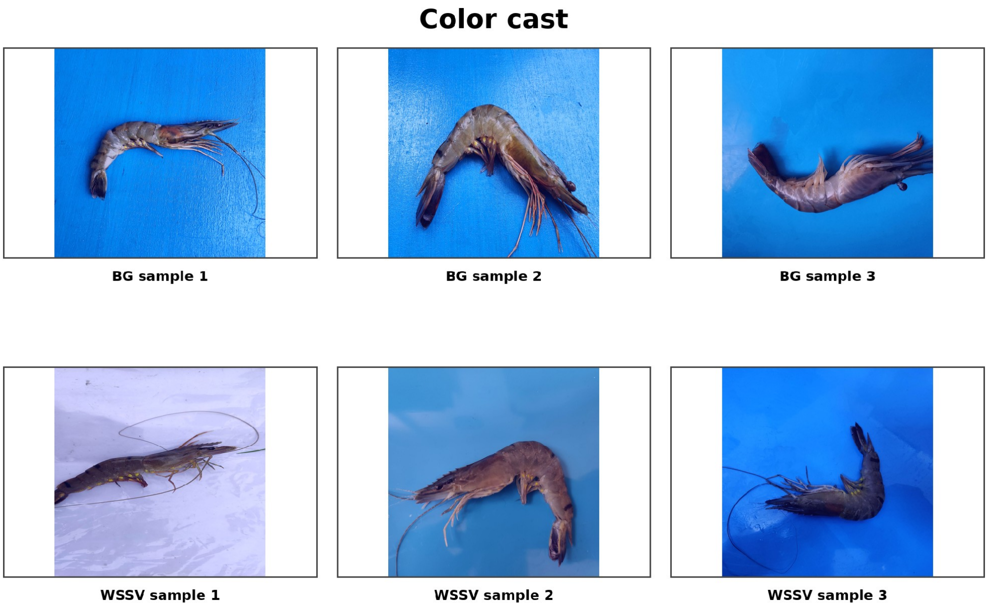

Figure 6.16: Representative color-cast segmentation examples for BG and WSSV. The original class and annotation semantics are retained; the panel is qualitative and does not quantify mask accuracy.

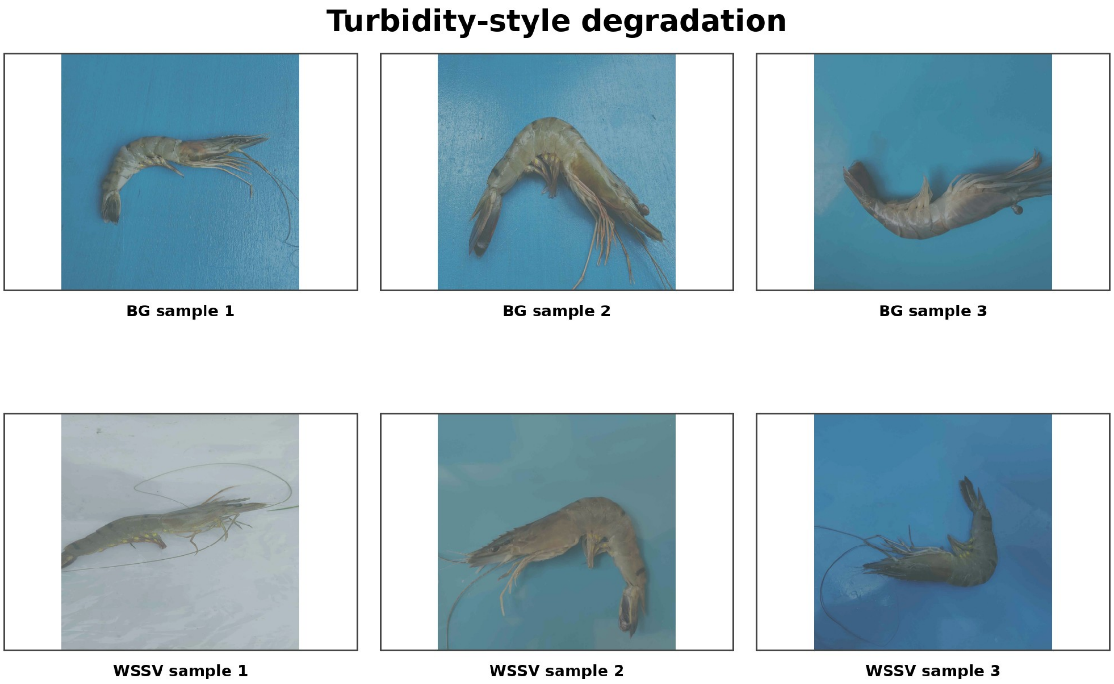

Figure 6.17: Representative turbidity-style segmentation examples for BG and WSSV. The original class and annotation semantics are retained; the panel is qualitative and does not quantify mask accuracy.

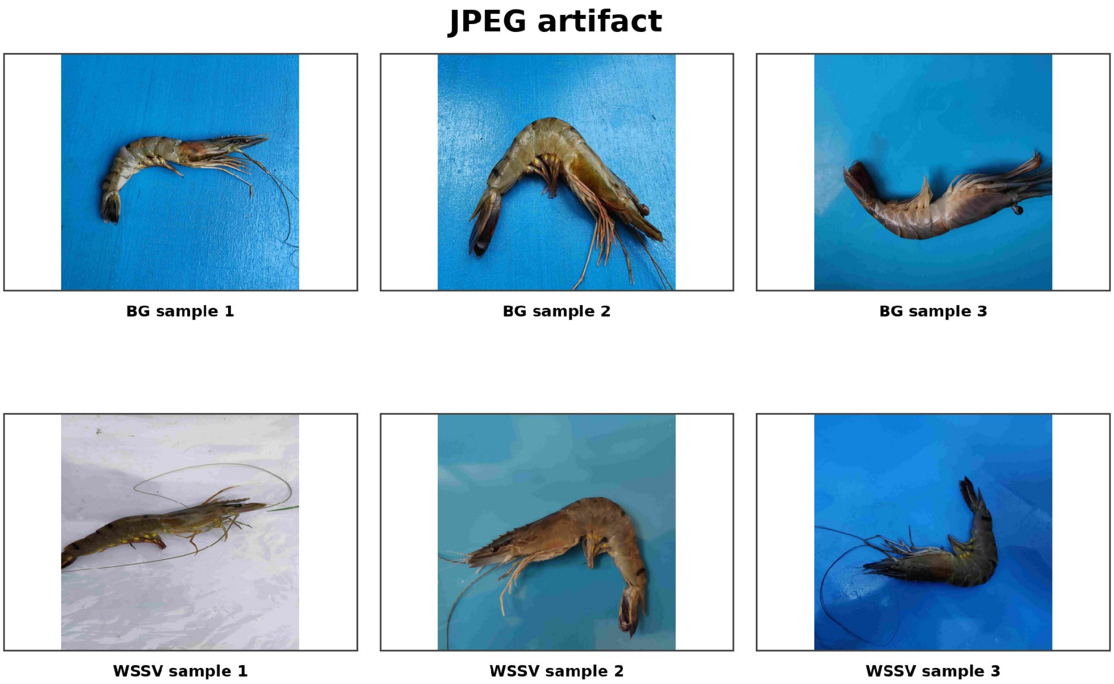

Figure 6.18: Representative JPEG-artifact segmentation examples for BG and WSSV. The original class and annotation semantics are retained; the panel is qualitative and does not quantify mask accuracy.

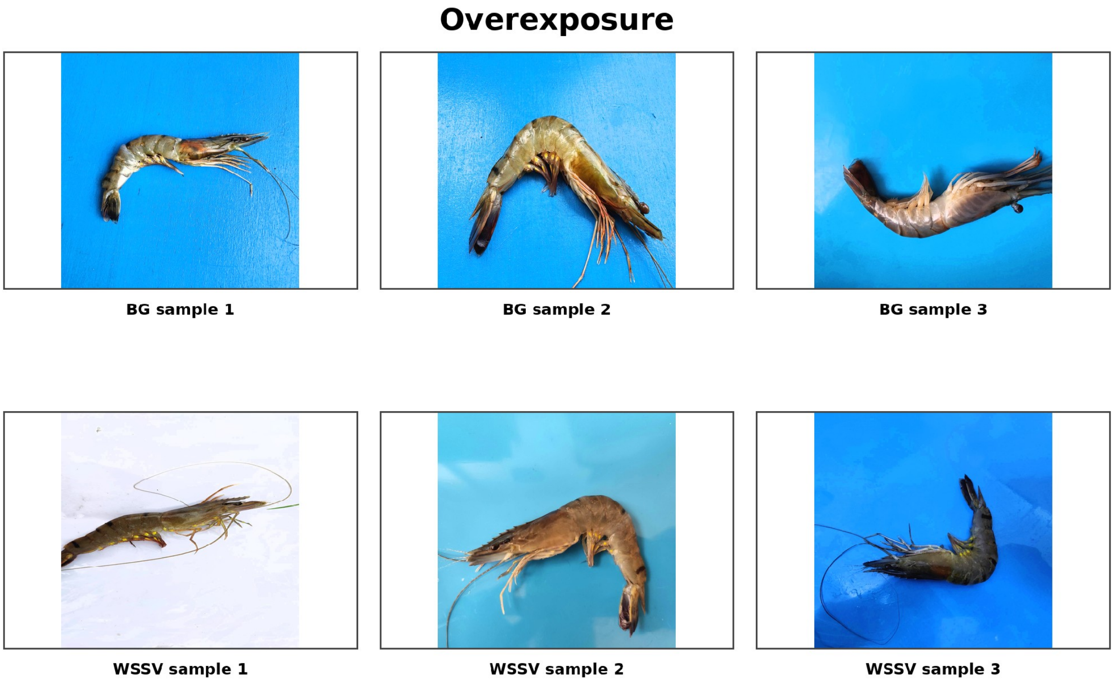

Figure 6.19: Representative overexposure segmentation examples for BG and WSSV. The original class and annotation semantics are retained; the panel is qualitative and does not quantify mask accuracy.

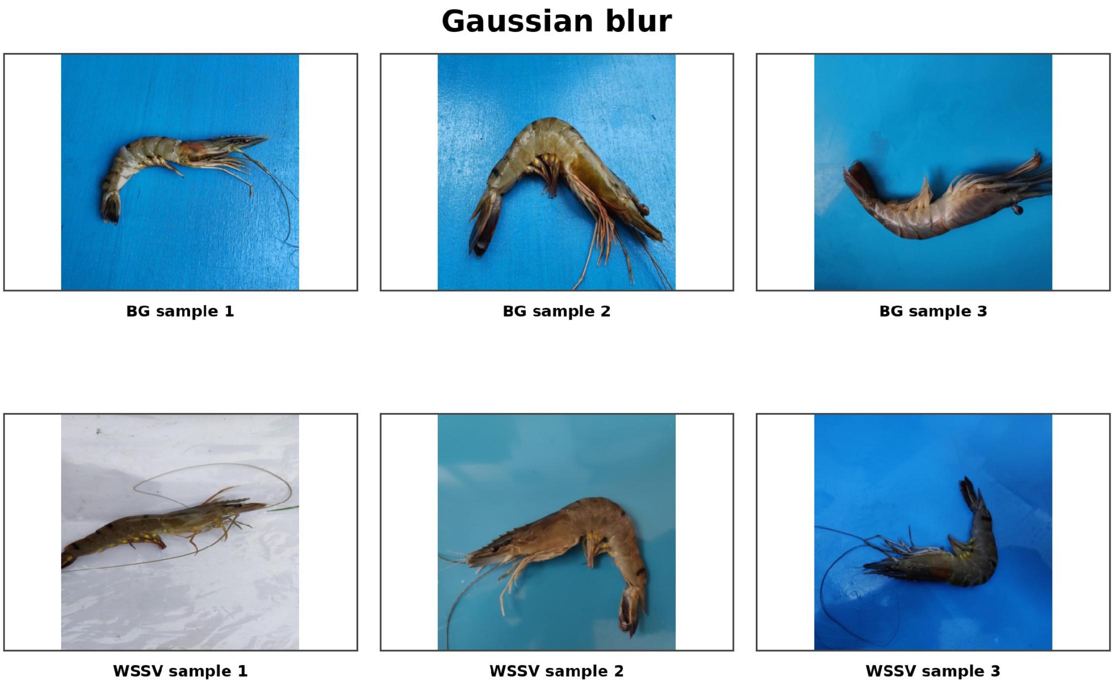

Figure 6.20: Representative Gaussian-blur segmentation examples for BG and WSSV. This retained source panel represents blur and is distinct from the Gaussian-noise metric row in Table 6.19; it is qualitative and does not quantify mask accuracy.

Figures 6.16–6.19 illustrate the first four metric-matched conditions in Table 6.19. Color cast and turbidity alter global appearance, JPEG artifacts introduce block and ringing structure, and overexposure removes contrast. The retained fifth image source, shown in Figure 6.20, is explicitly Gaussian blur, whereas the fifth quantitative row is Gaussian noise. Because these are distinct transformations, the blur panel is retained as separate qualitative evidence and is not used to illustrate or validate the Gaussian-noise metric.

Common segmentation failure mechanisms include missed small lesions, fragmented masks, mask spill beyond the visible symptom, Healthy false positives, and ambiguous overlap between BG and WSSV evidence. The Figure 6.21 panels below support qualitative inspection of these behaviors but are not the held-out test aggregate and do not quantify boundary error.

14. Mobile Deployment Observations

The revised mobile evaluation distinguishes *model compactness* from *on-device runtime*. The implemented Android workflow is a conditional two-model pipeline. Every image first passes through the FP32 classifier yolo26m_asl_ldam_simam_dcfr_fp32.tflite. If the classifier returns Healthy above the application threshold, the workflow stops after classification. If it returns BG, WSSV, or WSSV_BG above threshold, the application additionally invokes the FP16 instance-segmentation artifact yolo11n_seg_float16.tflite. Low-confidence outputs also terminate without segmentation. Therefore a single "mobile inference time" must not be interpreted as the cost of one fixed model path.

Table 6.20: Final conditional Android inference contract

Item	Implemented / retained evidence	Interpretation boundary
Stage 1 classifier	YOLO26m-cls + ASL-LDAM + SimAM-DCFR; FP32 TFLite; input $[1, 224, 224, 3]$; output $[1, 4]$	Runs for every image; ASL-LDAM is training-only
Dispatch rule	Healthy or below-threshold output: no segmentation; BG/WSSV/WSSV_BG above threshold: invoke segmentation	Predicted branch is an application control signal, not verified disease ground truth
Stage 2 segmentation	YOLO11n-seg + CA → SimAM at P3/P4/P5; FP16 TFLite, 5.65 MB	Runs conditionally for disease-triggered outputs
Execution scope	Android CPU path; four threads in the configured application	No iOS/iPhone test; no verified cross-device GPU/NNAPI comparison
Logged output	Predicted class, confidence, branch-dependent runtime, history record	Ground-truth fields are empty in the field runtime logs; no mobile accuracy is reported
Timing boundary	Application-recorded elapsed time for the operational pipeline	Not isolated pure network latency; may include preprocessing, interpreter/postprocessing, and application overhead

Table 6.21: **Final selected instance-segmentation model summary**

Item	Verified information
Final segmentation model	YOLO11n-seg + CA → SimAM at P3/P4/P5
Input	640 × 640 RGB
Parameters	2,854,718 (2.855M)
Complexity	Approximately 9.7 GFLOPs
Artifact size	best.pt approximately 6.1 MB; yolo11n_seg_float16.tflite 5.65 MB
Key localization results	Full-test mask mAP50 49.71%; labeled mask mAP50 55.07%; labeled mask mAP50-95 19.64%; full-test recall 49.70%
Notebook timing	9.3 ms/image; 107.5 FPS on the retained notebook GPU timing
Selection rationale	Highest labeled-test localization in the retained attention set with low model complexity and only a small parameter increase over the YOLO11n-seg reference

The 9.3 ms/image value in Table 6.21 is a notebook GPU measurement and is deliberately separated from the Android results below. A model can be compact in parameters and serialized size without achieving the same speed on every phone.

Table 6.22: **Android devices used for cross-device runtime profiling**

Member	Device	Android	SoC	CPU cores	RAM as reported	Storage
Tran Huu Nhan	vivo Y100	15	Snapdragon 685, up to 2.8 GHz	8	16 GB reported	128 GB
Le Thi Huynh Nhu	Samsung A32	13	MediaTek Helio G80, up to 2.00 GHz	8	6 + 6 GB	128 GB
Nguyen Van Phong	POCO M7 Pro 5G	16	MediaTek Dimensity 7025 Ultra	8	8 + 4 GB	256 GB

The RAM values are recorded as reported by the devices/users. Values such as 6+6 GB and 8+4 GB include vendor memory-extension configurations and should not be read as wholly physical RAM. GPU model fields were not retained in the supplied runtime records and are therefore not inferred.

Table 6.23: **Observed runtime of the conditional Android pipeline on three devices**

Device	All runs	No-seg branch, mean	Seg-triggered runs	Seg-triggered branch, mean	Overall mean
vivo Y100	21	14; 5.016 s	7	6.914 s	5.649 s
Samsung A32	21	2; 0.499 s[a]	19	2.155 s	1.997 s
POCO M7 Pro 5G	21	13; 1.860 s	8	2.641 s	2.157 s

[a] The two Samsung A32 no-seg observations were below-threshold short-circuit cases; no Healthy above-threshold record was present in that log. They are therefore not used for direct Healthy-branch hardware ranking. On the vivo Y100, eight Healthy above-threshold runs averaged 5.024 s; on the POCO M7 Pro 5G, ten Healthy above-threshold runs averaged 1.903 s.

All three Android devices used the same application build, the same FP32 classification artifact, and the same FP16 YOLO11n-seg artifact under the configured four-thread CPU execution path. Each member recorded 21 runs using manually uploaded real-world shrimp photographs. Image order was not standardized because the experiment was designed as a runtime-feasibility assessment rather than a controlled image-by-image hardware benchmark. Device-level means are therefore interpreted together with the observed branch composition and are not used as a strict hardware ranking.

The branch-specific results explain why disease-triggered cases can take longer than Healthy or other no-seg cases. Classification is executed first; an above-threshold disease output then dispatches the instance-segmentation model, whereas a Healthy or below-threshold result terminates after classification. On the vivo Y100, the segmentation-triggered branch averaged approximately 1.90 s longer than the aggregate no-seg branch; on the POCO M7 Pro 5G, the corresponding difference was approximately 0.78 s. The Samsung A32 log contains 19 segmentation-triggered records and only two below-threshold no-seg short-circuit cases, so its 0.499 s no-seg mean is not treated as a stable classification-only hardware estimate.

The three-device study evaluates runtime feasibility, not field diagnostic accuracy. The photographed shrimp did not receive veterinary, laboratory, or other expert disease ground truth; accordingly, the CSV ground-truth and correctness fields remain empty. Predicted class labels are used only to determine which application branch executes and are not treated as verified disease labels. No mobile accuracy, sensitivity, specificity, or disease-prevalence result is therefore calculated. iOS was not tested because no compatible iPhone/iOS device was available, and this remains a deployment limitation rather than an inference from the Android results.

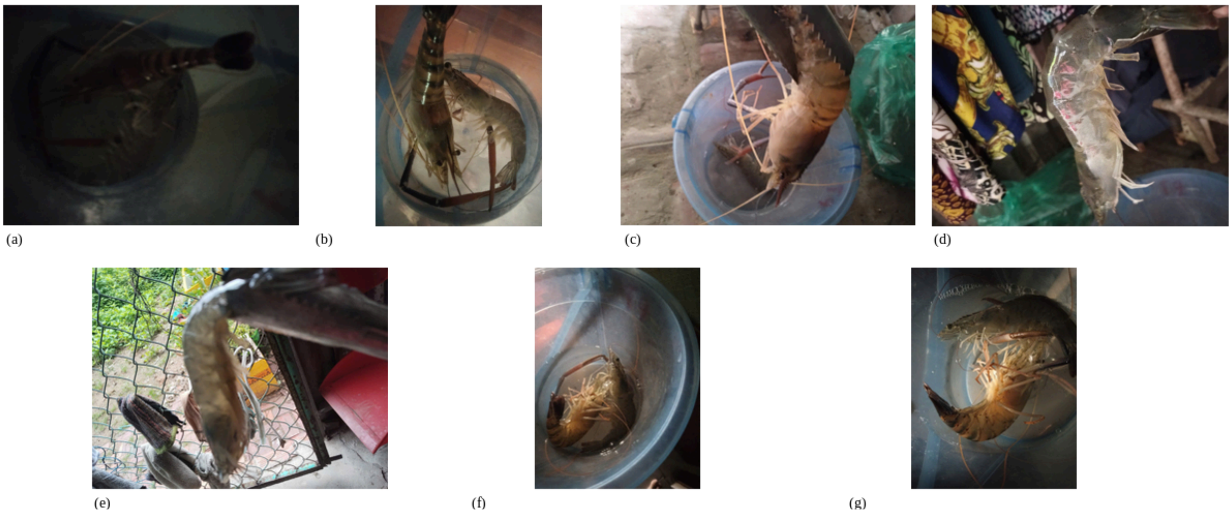

Figure 6.20A. Representative real-world shrimp photographs from the 21-image operational acquisition set captured using the three team-owned Android phones (vivo Y100, Samsung A32, and POCO M7 Pro 5G). The examples illustrate low illumination, uneven lighting, camera softness/blur, cluttered backgrounds, close-range framing, orientation changes, and varied viewpoints. These photographs support operational runtime testing only; they are not an expert-labeled field-accuracy benchmark.

The field photographs provide a closer operational analogue to the synthetic robustness motivation by exposing the application to naturally occurring capture variation from three Android cameras. They demonstrate that the deployed pipeline was exercised on non-dataset photographs under imperfect acquisition conditions. Because no veterinary or laboratory diagnosis was available, however, the photographs support deployment feasibility and capture-diversity discussion only; they do not establish disease-recognition accuracy or clinical generalization in real-world conditions.

15. Integrated Discussion

The primary classification result demonstrates an observed improvement under the recorded fixed-seed protocol. The 1.99-point Macro-F1 increase matters because the classes were imbalanced and visually overlapping classes require more than aggregate accuracy for interpretation. Kappa moved in the same direction as accuracy and Macro-F1, indicating that the primary improvement was not merely an increase in raw agreement. Nevertheless, the absence of a retained primary class-wise report prevents a complete account of which classes contributed most.

The additional-source experiments place an important boundary around the main result. The proposed configuration was much stronger than CE on EXT-3-Original but weaker than CE on SDI-4 + EXT-3-Original. This is evidence that loss design and feature recalibration interact with source composition, class priors, and visual domain. A method that improves one controlled dataset should not be described as uniformly superior across source combinations.

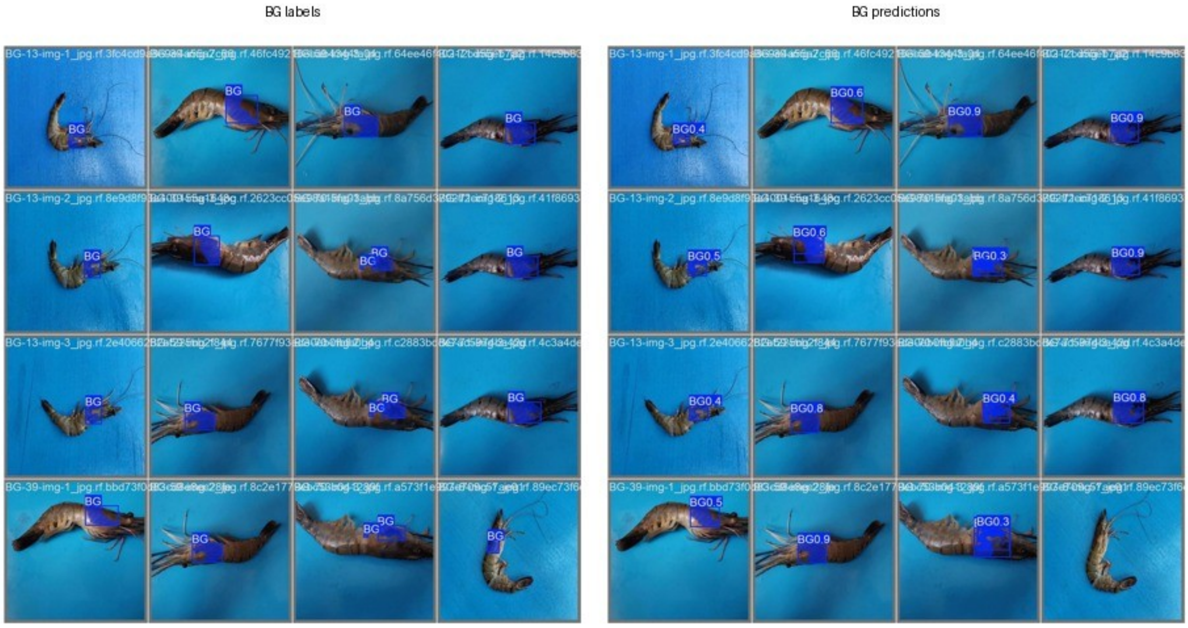

Figure 6.21(a). BG ground-truth labels and corresponding BG predictions.

Figure 6.21(b). Healthy empty-label negatives and corresponding disease-mask predictions.

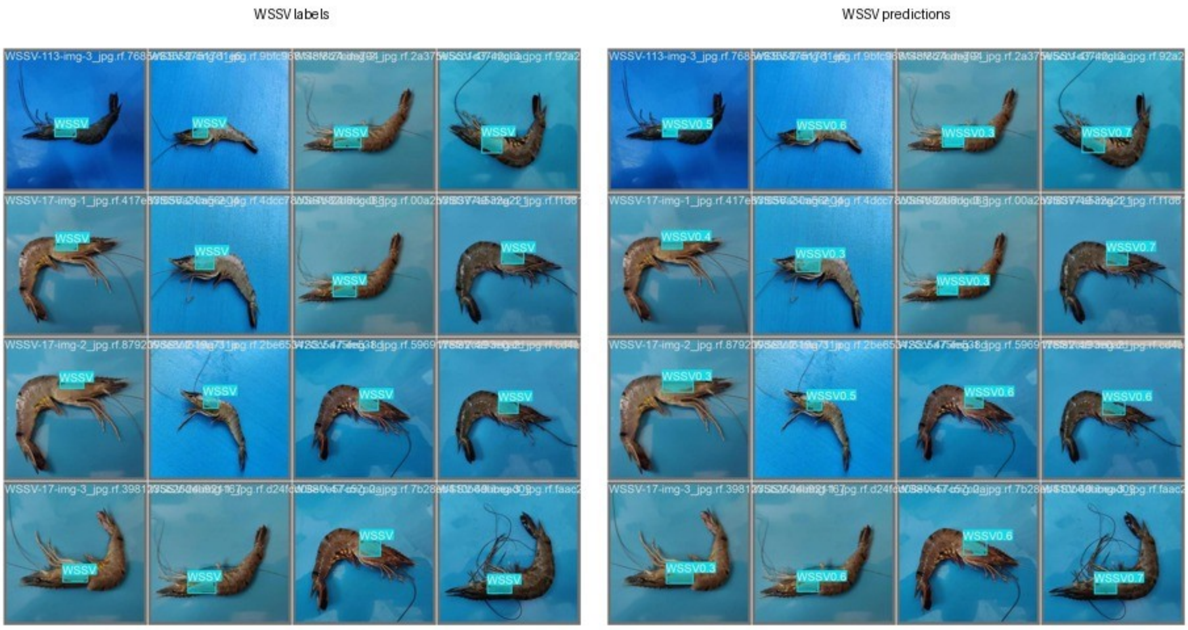

Figure 6.21(c). WSSV ground-truth labels and corresponding WSSV predictions.

Figure 6.21. Retained instance-segmentation ground-truth and prediction panels for BG, Healthy, and WSSV evaluation images. BG and WSSV rows compare project-authored disease-region ground-truth masks with predicted masks. Healthy rows are negative controls with no positive disease-mask target; every predicted BG or WSSV region in those rows is therefore a false positive.

The Healthy panel is a negative-control comparison. Healthy images contain no positive BG or WSSV disease-mask target, so every retained predicted disease region in the corresponding prediction panel is a false positive. This qualitative evidence explains why Healthy-negative diagnostics are reported separately from positive-mask average precision.

The corruption results support reduced sensitivity under five selected classification degradations, but the absolute impulse-noise score remained low. The segmentation corruption table was selected by positive gains and used a different set of transformations. These two branches therefore cannot be merged into one universal robustness claim. A unified benchmark with the same severity definitions and reporting policy would be needed for a system-level comparison.

XAI contributes a qualitative sanity check rather than a localization metric. Agreement among Grad-CAM++, HiResCAM, and EigenCAM over shrimp-body regions can reveal obvious background dependence, while disagreement shows that explanation is method-dependent. Neither a heatmap nor a segmentation mask establishes causality or veterinary validity.

The leakage analysis is one of the strongest validity findings. Random image splitting produced much larger mask mAP while nearly identical specimens crossed partitions. The grouped protocol reduced apparent performance but provided a more credible estimate for previously unseen filename-derived specimens. Lower grouped mAP is therefore more informative than higher leaky mAP for the intended validity question.

The attention results show metric-specific trade-offs rather than one dominant configuration. CA-to-SimAM led labeled localization and HScore, LKA-to-SimAM led full-test mAP50 and disease miss rate, and DPCA minimized Healthy false positives. Protocol differences prevent assigning these changes solely to attention design. CoTEGate + BoundaryLite is especially non-isolated because it changes both the P4 feature path and the training loss.

The Fourier experiments expose a similar trade-off. W1 substantially improved the strict no-augmentation control but amplified Healthy false positives and imposed a large preprocessing cost. Under strong conventional augmentation, W1 slightly reduced mAP50. The defensible conclusion is that Fourier preprocessing is condition-dependent and must be evaluated with both localization and negative-image guardrails.

The revised deployment evidence now covers three Android devices and the complete classifier-plus-conditional-segmentation control flow. The key distinction is branch-dependent runtime: Healthy and below-threshold outcomes stop after classification, whereas disease-triggered outputs execute the additional FP16 YOLO11n-seg pipeline. This makes a single device-average latency dependent on workload composition. The real-world photographs were captured using the three team cameras under varied acquisition conditions, but because no veterinary/laboratory ground truth was available, they support operational robustness discussion and runtime profiling only, not field accuracy. No iOS device was evaluated.

Relative to the literature reviewed earlier in the thesis, the contribution is not a claim of superiority over incomparable published percentages. The stronger contribution is the combination of a controlled classification reference, explicit imbalance-aware and feature-recalibration design, synthetic corruption testing, multi-source qualification, specimen-grouped segmentation, Healthy-negative diagnostics, attention/Fourier trade-off analysis, and a traceable mobile interface. Each claim remains bounded by its dataset, split, checkpoint, and artifact provenance.

16. Threats to Validity and Limitations

The classification evidence uses one fixed seed, so run-to-run variance and statistical significance are unknown. Test-partition Macro-F1 was inspected during candidate screening, weakening model-family ranking independence. Complete primary classwise predictions and a primary confusion matrix were not retained, which limits exact error attribution. Dataset coverage remains constrained by public sources, class imbalance, acquisition bias, and background removal. The mapped EXT-3-Original and SDI-4 + EXT-3-Original experiments involved retraining and are not direct cross-dataset inference by the original checkpoint.

Segmentation specimen identities were inferred from filenames rather than independently verified metadata. One annotator produced the masks without inter-annotator agreement or expert boundary review, and deterministic single-label overlap handling simplified ambiguous tissue. Attention runs differed in augmentation, epoch budget, early stopping, software version, placement, and auxiliary loss; the combined extension additionally used a mixed grouped–stratified split. These results are comparative records rather than a strict one-factor ablation.

Robustness evidence covers selected synthetic conditions only, and classification and segmentation used different corruption families and reporting rules. The segmentation table contains only the five largest positive gains. XAI remains qualitative. Fourier preprocessing introduces latency and false-positive trade-offs, and mobile evidence is limited to a prototype device observation without formal farmer testing, complete delegate evaluation, or proof of final segmentation deployment. The system supports preliminary screening and is not a clinical or veterinary diagnostic method.

The revised mobile evidence improves device coverage from one bounded observation to three Android phones, but it remains an observational runtime study rather than a controlled cross-device benchmark. Workload composition differs across the three logs, the application timer is not decomposed into pure network stages, and no iOS device was available. The real-world photographs were captured with the three team cameras under varied acquisition conditions, but they do not have expert or laboratory disease ground truth; consequently they cannot support a field-accuracy claim.

17. Recommendations

Future classification work should use validation-only model selection, retain an untouched final test partition, and run at least three to five seeds for the CE and proposed configurations. Frozen prediction files should support paired analysis, classwise reports, count and normalized confusion matrices, calibration, confidence-based failure analysis, and reproducible hashes for datasets, splits, checkpoints, environments, and conversions.

The segmentation dataset should receive independent multi-annotator review with mask-IoU or boundary-agreement analysis. Specimen identities should be strengthened through source metadata and duplicate detection, while external testing should preserve specimen, farm, and source groups. A unified corruption benchmark should apply common families, severity levels, aggregation rules, and full positive-and-negative reporting to both tasks.

The Council's conditional request for a U-Net comparison was not converted into an unverified numerical row. The original/standard U-Net formulation is a semantic pixel-classification architecture and does not inherently separate multiple instances, whereas the present task is evaluated as two-class instance segmentation with instance-level mAP and Healthy-negative diagnostics. Dice or semantic IoU from a standard U-Net cannot be directly ranked against YOLO instance mAP without a matched conversion/evaluation protocol. Instance-aware U-Net-derived variants exist in the literature, but no reproducible implementation was completed under the same grouped-specimen manifest, approximately 15M-or-lower model budget, training budget, annotation semantics, and metric definitions before the revision deadline. A fair future benchmark should use one shared manifest and report common semantic metrics (Dice/IoU/mIoU) plus an explicitly defined instance-extraction protocol if instance-level comparison is required.

The endpoint comparison with RTMDet-Ins-S is already retained in Table 6.9B. If verified RTMDet train/validation logs become available, the next revision should add a matched YOLO11n-seg versus RTMDet-Ins-S optimization-trajectory figure at this point in the segmentation discussion. Until those logs are available, no train/validation curve should be reconstructed from final metrics.

Deployment evaluation should compare quantized and FP32 artifacts under one interface contract and separate decode, preprocessing, initialization, inference, postprocessing, and UI timing. Cross-device CPU and supported-delegate profiling should record warm-up, thread count, hashes, app version, and device identifiers. The final segmentation checkpoint requires explicit TFLite/LiteRT export and an end-to-end mask regression suite. A controlled user study should verify that confidence, masks, limitations, and escalation guidance are understood without being interpreted as diagnosis.

18. Personal Reflection

The principal methodological lesson was that a large metric is not sufficient evidence. Baseline reproduction, split integrity, artifact provenance, negative-image behavior, and protocol comparability materially changed the interpretation of the results. The specimen-grouped analysis was particularly important because it showed that a convenient split could overstate localization performance.

Model research and system integration remained distinct: a controlled GPU gain did not imply mobile speed, and higher segmentation mAP could coincide with more Healthy false positives. Future work should preserve exact protocols, expose trade-offs, and keep claims bounded.